

Birthday Cutoffs and Hereditary Sets

The set-theoretic content of bounded surreal arithmetic:
bi-interpretations, elementary inclusions, axiom spectra,
outer models, and surcomplex orientation

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Abstract

Expand the ordered field of surreal numbers of birthday below an ordinal λ by its birthday function. The global connection of this kind of structure with set theory is the subject of a bi-interpretation theorem announced by Chen, Hamkins and Yang. This report, merged from five manuscripts that prove overlapping versions of one local theorem, asks *which set universe is recovered from a particular birthday cutoff, and what that recovery implies*.

For an epsilon number λ let $\kappa(\lambda)$ be the least cardinal not below λ . One parameter-free interpretation recovers exactly $H_{\kappa(\lambda)}$. At cardinal cutoffs, singular ones included, this is a uniform bi-interpretation with H_λ ; at noncardinal cutoffs it is a bi-interpretation with $(H_{\kappa(\lambda)}, \in, \lambda)$, and the cutoff is recovered by a definable collector. Thus the field of birthdays below ε_0 already recovers all hereditarily countable sets. Small arithmetic certificates justify the reverse interpretation without Replacement in H_κ , and a second route codes graphs modulo bisimulation. The forward interpretation extends to the ring cutoffs, including ω .

Consequences within one universe: for epsilon numbers $\lambda < \eta$ the inclusion of birthday structures is elementary exactly when both are cardinals and $H_\lambda \prec H_\eta$; elementary embeddings correspond to those of the hereditary universes; one sentence detects noncardinal cutoffs and another the strong-limit cardinals; birthday eternity is Replacement at cardinal cutoffs, and with fiber packing it selects the worldly cardinals; in ZF the graph coding reaches exactly the sets with well-orderable transitive closure. Across universes: old arithmetic is absolute, a bounded structure is unchanged exactly when no bounded subset is added, the first new birthday is the least ordinal with a new subset and is a ground-model cardinal, explicit formulas detect new subsets, and, assuming a measurable cardinal, Prikry forcing leaves the enriched field unchanged while singularizing its cutoff. Under GBC the full structure has no nontrivial elementary self-embedding. For surcomplex numbers with conjugation and coordinate birthday, each real elementary embedding has exactly two lifts, and one nonreal parameter, but no set of real parameters, defines an orientation.

Built from five manuscripts written independently in relation to the `VladimirReshetnikov/Surreal` project,
all pinned to commit `4f26455`, and merged after placement commit `d4e71b7`.

An AI-assisted research draft with written proofs and finite regression checks; not refereed;
not a claim of any Lean formalization. The global bi-interpretation is prior announced work of Chen–Hamkins–Yang.

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Part I

The cutoff theorem

1 The question, the results, and their provenance

The field structure of the surreal numbers and their construction history carry very different information. Real-closed-field quantifier elimination makes all inclusions between real closed subfields elementary in the ordered field language. A birthday function remembers much more: together with the order it reads individual sign coordinates, and thereby encodes subsets, relations and well-founded membership graphs. The issue addressed here is not merely whether this coding is possible globally, but how its strength changes when only birthdays below one ordinal are allowed, and what the resulting dictionary says about elementary inclusions, set-existence axioms, outer models of set theory, and the surcomplex numbers.

1.1 What is already known, and credited

The global theorem is prior work. Chen, Hamkins, and Yang announced that the surreal ordered field with birthday order is bi-interpretable with the set-theoretic universe, and described a first-order theory of surreal arithmetic intended to be bi-interpretable with ZFC. Hamkins’s September and November 2025 slides give the global coding by pointed well-founded extensional relations, together with birthday-growth principles; the March 2026 workshop announcement and slides repeat the global theorem and label the axiomatic work as still being checked [6, 7, 8]. This report does *not* claim that theorem, its pointed-graph coding idea, or the association of birthday growth with set-existence axioms as new. The collection’s **foundations-and-computation/foundations** report already records the March 2026 announcement, at slide level and with its status boundary, in its subsection **found:sub:announcement**, and lists it among its non-claims as work in progress that is not a proved dependency of that report. Of the five manuscripts merged here, sources 02, 06 and 09 credit the announcement prominently. Sources 05 and 07 do not cite it; source 07 presented the full-class bi-interpretation as a proposed new contribution. That presentation is corrected here: [theorem 18.1](#) states the announced theorem with its credit, and the argument printed there, taken from source 07, is a written proof of the announced theorem, not a new result.

Other prior inputs. If λ is an epsilon number, $\mathbf{No}_{<\lambda}$ is a real closed subfield. The original van den Dries–Ehrlich paper must be used together with its erratum [3, 4]; Bournez–Guilmant state the corrected ring, field and real-closedness results in their Theorems 1.1–1.2 [5]. The collection’s foundations report states the same classical fact in its subsection **found:sub:cutoffs**, including that ε_0 is already an epsilon number, so that no large cardinal is needed. The sign construction and recursive arithmetic are classical [1, 2]. Mostowski collapse, well-founded recursion, hereditary-size universes and forcing are standard [13]; quantifier elimination for real closed and algebraically closed fields is standard [14, 15]. The general obstruction to parameter-free bi-interpretation caused by complex conjugation, and the role of a named imaginary unit, were explained by Jeřábek in 2021 [9]; every use of that obstruction below ([section 19](#)) is credited to him. Bi-interpretation methodology follows Freire–Hamkins [10]. The Prikry forcing facts are imported from Poveda–Rinot–Sinapova [11], and the class form of Kunen’s inconsistency from Hamkins–Kirmayer–Perlmutter [12]. Rangel–Mariano’s algebraic set theory of surreal numbers and Bertram’s realization of numbers as sets of

ordinals are separate programs with different languages and aims; nothing here settles or refutes them [16, 17].

1.2 The principal statements

We use $\text{TC}(\{x\})$ for the transitive closure containing x and all its iterated members, and write

$$H_\kappa = \{x : |\text{TC}(\{x\})| < \kappa\}.$$

For an epsilon number λ (an ordinal with $\omega^\lambda = \lambda$ in ordinary ordinal exponentiation) put

$$\mathcal{B}_\lambda = (\mathbf{No}_{<\lambda}; 0, 1, +, -, \cdot, <, \mathbf{b}),$$

where $\mathbf{b}(x)$ is the birthday of x , regarded as the all-plus surreal ordinal of that length. “Uniform” means that fixed finite formulas work at every cutoff in the stated class, not that a separate language is chosen for each cardinal. The source numbers in brackets are explained in [section 1.3](#).

Theorem 1.1 (Cardinal rounding and cutoff recovery; [theorem 6.3](#)). *Let λ be an epsilon number and let*

$$\kappa(\lambda) = \min\{\kappa : \kappa \text{ is a cardinal and } \lambda \leq \kappa\}.$$

There is one parameter-free interpretation $\mathcal{I}$ such that

$$\mathcal{I}(\mathcal{B}_\lambda) \cong (H_{\kappa(\lambda)}, \in).$$

If λ is a cardinal, $\mathcal{B}_\lambda$ and $(H_\lambda, \in)$ are uniformly bi-interpretable. If λ is not a cardinal, $\mathcal{B}_\lambda$ and $(H_{\kappa(\lambda)}, \in, \lambda)$ are uniformly bi-interpretable, and the named ordinal λ is definable, as an interpreted set, in $\mathcal{B}_\lambda$ itself.

This is the most general form in the cluster, and it is source 09’s. The other four sources prove cases of it with their own codes:

- source 02: the forward interpretation onto $H_{\kappa(\lambda)}$ at every epsilon number, one way only (it explicitly claims no reverse interpretation);
- source 05: the bi-interpretation at uncountable *regular* cardinals, with graphs modulo bisimulation ([section 7](#));
- sources 06 and 07: the bi-interpretation at every uncountable cardinal, singular ones included; source 07 also proves the forward interpretation onto $H_{\kappa(\lambda)}$ at every epsilon number, which it calls a *cardinality plateau* and states forward only.

This is a cardinal-rounding statement, not a rank-cutoff statement: for example $\kappa(\varepsilon_0) = \omega_1$, whereas V_{ε_0} is not the recovered universe. At a singular cardinal κ the recovered structure is still H_κ ; no regularity assumption is hidden in the theorem.

Theorem 1.2 (Elementary-inclusion spectrum; [theorem 9.3](#)). *For epsilon numbers $\lambda < \eta$,*

$$\mathcal{B}_\lambda \prec \mathcal{B}_\eta \iff \lambda, \eta \text{ are cardinals and } H_\lambda \prec H_\eta.$$

The ordered-field reducts, in contrast, always form an elementary inclusion.

Source 09 proves the equivalence at all epsilon numbers. Source 02 proves the forward implication and left the converse as its first open question; that question is answered by source 09 ([remark 9.4](#)). Sources 05 (regular cardinals), 06 and 07 (uncountable cardinals) prove the equivalence between cardinal cutoffs, and 05 and 06 add the correspondence of elementary embeddings ([theorem 9.10](#)).

Theorem 1.3 (Growth spectrum; [theorem 11.2](#)). *For an epsilon-number cutoff, birthday eternity Et fails at every noncardinal cutoff. At an uncountable cardinal κ ,*

$$\mathcal{B}_\kappa \models \text{Et} \iff H_\kappa \models \text{Replacement}.$$

Fiber packing Pack, and also the translated Power Set sentence $\text{Pow}^\mathcal{I}$, hold exactly at the uncountable strong-limit cardinal cutoffs. Consequently

$$\mathcal{B}_\lambda \models \text{Et} + \text{Pack} \iff \lambda \text{ is a cardinal and } V_\lambda \models \text{ZFC}.$$

The eternity, packing and worldly statements are source 09's. The translated Power Set sentence is source 02's, at every epsilon number, and source 06's at uncountable cardinals ([theorem 10.7](#)).

Theorem 1.4 (Extension rigidity at a strong-limit cutoff; [theorem 16.4](#)). *Let $M \subseteq N$ be transitive models of ZFC with the same ordinals, and let κ be an uncountable cardinal in both, strong limit in M . Then*

$$\mathcal{B}_\kappa^M \prec \mathcal{B}_\kappa^N \iff \mathcal{B}_\kappa^M = \mathcal{B}_\kappa^N \iff \mathcal{P}^M(\alpha) = \mathcal{P}^N(\alpha) \quad (\alpha < \kappa).$$

Without the strong-limit hypothesis, a new subset of $\alpha < \kappa$ still prevents elementarity whenever $(2^{|\alpha|})^M < \kappa$.

The detector is sharp for the old-power-set *parameter* ([proposition 16.2](#)), not a claim that every other possible detector needs that much space. The report also proves:

- the forward interpretation at ring cutoffs, including $\mathbf{No}_{<\omega}$ onto the hereditarily finite sets ([proposition 8.1](#), source 02);
- one sentence true exactly at noncardinal epsilon cutoffs ([theorem 6.2](#), sources 02 and 09);
- a separation of $\mathcal{B}_{\omega^+}$ from $\mathcal{B}_{\omega^+}$ by one Replacement instance ([theorem 11.5](#), source 06);
- the exact range of the graph coding over ZF without Choice ([theorem 13.2](#), source 06);
- the transfer of elementary equivalence, the interpretation of full second-order arithmetic, the independence property of the bit formula, an omitted parameter-free type, and nondefinability of birthday in the field reduct ([section 12](#), sources 05, 06, 07);
- the exact preservation criterion across outer models, the first new birthday, three explicit new-subset detectors, and the full-class statement $\mathcal{B}^M \prec \mathcal{B}^N \iff M = N$ ([Part III](#), sources 02, 05, 06, 07, 09);
- conditional on a measurable cardinal, an unchanged enriched field whose cutoff changes cofinality, the preserved $H_\kappa = V_\kappa^M$, and the exact onset of new elements above the cutoff ([section 17](#), sources 02 and 06);
- under GBC, no nontrivial class elementary self-embedding of the full birthday structure, by transfer of Kunen's inconsistency ([theorem 18.3](#), source 07);
- for surcomplex numbers with conjugation and coordinate birthday: exactly two elementary lifts of each real elementary embedding (source 06), the exact parameter threshold for a definable orientation (source 06), the mutual-interpretation versus bi-interpretation distinction (sources 05, 06, 09, after Jeřábek), and, under GBC, the elementary self-embeddings of the full structure (source 07) ([section 19](#)).

1.3 Five manuscripts, one report

This report is built from five manuscripts, written independently on September 22, 2026, all pinned to repository commit `4f2645599121fa872c7104995e47f86f0382351f` and placed in the collection by commit `d4e71b7`. Their numbers are those of the placement batch and are local to this directory.

Source	Manuscript	Contribution to this report
09	<i>The Set-Theoretic Content of Bounded Surreal Arithmetic: hereditary-size universes, coherent bi-interpretations, axiom spectra, and extension rigidity</i>	The base. Its text is the spine of Part I (sections 2 to 6), the elementary-inclusion classification (section 9), the eternity, packing and worldly calibration (section 10), the cross-model coherence, detector and extension rigidity (section 16), the orientation obstruction (theorem 19.4), and most of section 20. Files prefixed <code>09-bounded-arithmetic-</code> .
02	<i>What a Birthday Cutoff Remembers: hereditary set universes, elementary obstructions, and forcing invisibility</i>	The one-way forward theorem at all epsilon numbers (a special case of the base), ring cutoffs (section 8), the collector sentence (printed once with 09's), the translated Power Set sentence at every epsilon number, the preservation criterion at epsilon cutoffs and its countable corollary, the Prikry invisibility theorem with its preserved H_κ and the onset above the cutoff (section 17), the concrete example in Appendix C. Files prefixed <code>02-birthday-cutoffs-</code> , and its source audit <code>02-birthday-cutoffs-SOURCE_AUDIT.md</code> .
05	<i>Birthdays Recover Sets: bi-interpretability, elementary embeddings, and forcing in surreal and surcomplex fields</i>	The second route by codes modulo bisimulation (section 7), the explicit embedding formula and critical-point corollary, the separation of $\mathcal{B}_{\omega_1}$ and $\mathcal{B}_{\omega_2}$, nondefinability of birthday, the explicit missing-row detector New, the full-class statement $\mathcal{B}^M \prec \mathcal{B}^N \iff M = N$, the first new birthday for arbitrary same-ordinal outer models, the countably closed example, and the surcomplex forcing contrast. Files prefixed <code>05-birthdays-recover-sets-</code> .
06	<i>What Birthday-Enriched Surreal Fields Remember: hereditary-size interpretations, choice, forcing, and surcomplex orientation</i>	The bi-interpretation at singular cardinals (printed once with the base), the second route by prefix-pair arithmetic tables (section 3.4), definitional equivalence with $(\mathbf{No}_{<\kappa}; <, \mathbf{b}, G_\varpi)$, transfer of elementary equivalence, the $\mathfrak{I}_\omega$ Replacement instance, the ZF range theorem, the four-way preservation criterion, the Prikry corollaries, the two complex lifts, and the orientation threshold. Files prefixed <code>06-hereditary-size-orientation-</code> .

Source	Manuscript	Contribution to this report
07	<i>The Birthday Expansion of the Surreal Field: recovering set theory, detecting forcing, and rigidity of surcomplex embeddings</i>	The written proof of the announced full-class theorem (theorem 18.1), the radix pairing, the ε_0 localization (printed once as a case of the base), recovery of models of ZFC, stage agreement at every ordinal, the two-parameter outer-model witness, the independence property and omitted type, the o-minimal nondefinability, Kunen transfer (theorem 18.3) and the surcomplex elementary self-embeddings. Files prefixed 07-birthday-expansion-.

Why one report. All five prove the same spine: the ordered field of surreals below a cutoff, expanded by birthday, interprets the hereditary sets H_κ by sign bits read from order and birthday, a bounded ordinal pairing polynomial, and pointed well-founded relation codes, and in most versions is bi-interpretable with them. They differ in the cutoff class and in whether the result is one-way or two-way. Source 09 has the widest cutoff class (all epsilon numbers, which include all uncountable cardinals by [lemma 2.2](#)) and the strongest conclusion (two-way, with the correct pointed target), so it is the base.

Printed once. The prefix and bit formulas (all five), the bounded pairing (all five, in four variants compared in [remark 4.4](#)), the rooted collapse and exact range (02, 06, 07, 09), the arithmetic certificates (05, 06, 07, 09), the bi-interpretation at cardinal cutoffs (05 regular; 06, 07, 09 all), the ε_0 case (02, 07, 09), the collector sentence (02, 09), the elementary-inclusion classification (02 one direction; 05, 06, 07, 09), the translated Power Set threshold (02, 06, with 09's version inside H_κ), absoluteness of old arithmetic (02, 05, 07, 09), the preservation criterion (02, 06, 07, 09), the first new birthday (02, 05, 07), the outer-model non-elementarity (05, 07, 09), the Prikry invisibility (02, 06), rigidity of the bounded structures (05, 06, 07, 09), the surcomplex real-axis recovery (all five) and the two-element automorphism group (05, 06, 07, 09). Each is printed once and credited to every source that proves it.

Kept twice, as different proofs. The reverse arithmetic has two routes: source 09's option-graph certificates built from finite signed words ([theorem 3.3](#)), and the prefix-pair tables of sources 05, 06 and 07 ([section 3.4](#)), whose single- μ size bound is source 06's argument at singular cutoffs. The coding of sets has two routes: source 09's extensional rooted codes with isomorphism as equality ([section 5](#)), and source 05's codes modulo bisimulation ([section 7](#)). The first new birthday has two existence arguments: the new-subset lemma of 05 and 07 for any outer model, and the generic filter of 02 for set forcing ([remark 15.5](#)). The new-subset detectors are three different formulas ([section 16](#)).

Where the merge had to choose. (i) Hypotheses: whenever a result is proved by several sources under different hypotheses, the weakest proved hypothesis is printed, and the proof printed is one that uses only it; where this attaches a source's argument to a weaker hypothesis than the source stated, the fact is said at the point of use ([remarks 3.7](#) and [16.7](#)). (ii) The first new birthday is stated for arbitrary same-ordinal transitive outer models (05, 07), with 02's set-forcing

theorem as a special case. (iii) The full-class bi-interpretation is printed as the announced theorem of Chen–Hamkins–Yang, not as a result of source 07. (iv) Symbols were renamed as listed in [section 1.4](#).

1.4 Conventions and renamed symbols

Convention 1.5 (One meaning per symbol). The following conventions hold throughout, including in material taken from sources that used other letters.

Notion	Here	In the sources
Birthday of x , as a surreal ordinal	$\mathbf{b}(x)$	$\mathbf{b}$ (02); b (05, 07, 09); b for the ordinal and $\mathbf{b}$ for its surreal (06)
Cutoff carrier	$\mathbf{No}_{<\lambda}$	F_λ (02), S_κ (05, 06), $\mathbf{No}_{<\kappa}$ (07), K_λ (09)
Birthday structure	$\mathcal{B}_\lambda$; full class $\mathcal{B}$	$\mathfrak{B}_\lambda$ (02), $\mathcal{S}_\kappa$ (05), $\mathcal{A}_\kappa$ (06), $\mathcal{B}_\kappa$ (07, 09)
Cardinal ceiling	$\kappa(\lambda)$	$\theta(\lambda)$, the least <i>uncountable</i> cardinal $\geq \lambda$ (07); equal for epsilon λ
Pairing length	$\Lambda(\theta) = \theta^2 + \theta$	B_θ (02); $L(a, d)$ (05); $\tau(\delta)$ (06); q^2 (07)
Code	$C = (\delta, r, t)$: domain, relation word, root	(θ, d, e, r) (02); (λ, c, r) with λ a node count (05); (δ, e, r) (06); (q, d, e, r) (07)
Interpreted $=, \in$	$\sim, \in_{\mathcal{I}}$	$\mathbf{Eq}, \mathbf{Mem}$ (02); $\equiv, \in^*$ (06); $\simeq, \in_*$ (07)
Ordinal code	$\mathbf{OC}(a)$	$\mathbf{OC}$ (02), $\mathbf{Sync}$ (05), $\mathbf{OrdCode}$ (06), $\mathbf{OCode}$ (07), $\mathbf{O}(a)$ (09)
Collector	$\mathbf{Coll}(C)$; sentence $\exists C \mathbf{Coll}(C)$	$\mathbf{CollectOrd}$ (02), $\mathbf{Coll}$ (09)
Replacement instance at $\beth_\omega$	$\mathbf{Rep}_{\beth}$	$\mathbf{Collect}_{\beth}$ (06)
Translated Power Set	$\mathbf{Pow}^{\mathcal{I}}$	$\mathbf{Power}$ (02), $\mathbf{Pow}^\sharp$ (06)
Eternity, fiber packing	$\mathbf{Et}, \mathbf{Pack}$	$\mathbf{E}, \mathbf{P}$ (09)
Conjugation	c	c (02, 06); overline (05, 07); σ (09)
Coordinate birthday	$\mathbf{cb}(x + iy) = \max(\mathbf{b}x, \mathbf{b}y)$	β (02, 05, 09); B (06, 07)
Surcomplex structures	$\mathcal{C}_\lambda$ (no i named), $\mathcal{C}_\lambda^i$ (i named); full $\mathcal{C}, \mathcal{C}^i$	$\mathfrak{K}_\lambda$ with i (02); $\mathcal{C}_\kappa, \mathcal{C}_\kappa^-$ (05); $\mathcal{K}_\kappa$ (06); $\mathcal{C}_b, \mathcal{C}_b^i$ (07); $\mathcal{C}_\lambda^\pm$ (09)
First new birthday	$\beta = \beta(M, N)$	δ (02, 05, 07)
New-subset formula of 05	$\mathbf{New}(m, \theta, \nu)$	$\mathbf{New}(c, \theta, \lambda)$, λ an enumeration length (05)
Eternity bound	ϱ	β (09)

Three collisions with the collection’s shared notation are resolved as follows ([docs/NOTATION.md](#)). The collection reserves F_Γ and K_Γ for the real and complex Hahn workspaces and K for a set-sized *complex* field; sources 02 and 09 used F_λ and K_λ for birthday cutoffs (09’s K_λ a *real* field), so this report writes $\mathbf{No}_{<\lambda}$ and $\mathbf{No}_{<\lambda}[i]$ instead. The collection’s H_α is a subgroup of a value group (bounded multiples of α); here H_κ is always the universe of sets of hereditary size below κ , and no value group occurs. The collection’s $\mathcal{O}$ is a valuation ring, hence the ordinal code is written

OC. The letter β is reserved for the first new birthday of an outer model, as in the sibling report [foundations-and-computation/surreal-fields-across-universes](#); that report uses δ for the first new *ordinal-sequence length*, a notion not named here, while here δ is only the domain ordinal of a code. Sources 02, 05 and 07 wrote δ for the first new birthday.

Two further words have two meanings in the sources. “Collect” named a sentence collecting the literal ordinals in source 02 and a Replacement instance in source 06; the two are $\exists C \text{ Coll}(C)$ and $\text{Rep}_{\sqsupset}$ here. A sans-serif P was fiber packing in source 09 and Power Set in sources 02 and 06; they are Pack and $\text{Pow}^{\mathcal{I}}$ here, and Pack is *not* the Power Set axiom. Birthday precedence and prefix simplicity are different relations (`found:rem:simplicity` of the foundations report); prefix is defined below from order and birthday, and is never identified with precedence.

1.5 Relation to the collection and corrected repository statements

Foundations. The foundations report records the Chen–Hamkins–Yang announcement (`found:sub:announcement`), the epsilon-number cutoffs (`found:sub:cutoffs`) and the distinction between simplicity and birthday precedence (`found:rem:simplicity`). Its inventory notes that the field language has no birthday symbol; [proposition 12.5](#) supplies the standard nondefinability proof. None of the interpretation theorems of this report was in the collection before it.

Surcomplex automorphisms. The report [surcomplex/surcomplex-field-automorphisms](#) proves $\text{Aut}(K/F) = \{\text{id}, c\}$ for $K = F(i)$ with F real closed (`saut:thm:axis`) and, in its section `saut:sec:exponential`, subsection “Simplicity restores the canonical surreal numbers”, that a bijection preserving numerical order and prefix simplicity is the identity, so that a real-axis-preserving automorphism whose real restriction preserves simplicity is the identity or conjugation. Since order and birthday define prefix ([lemma 4.1](#)), the bounded automorphism statements of sources 05, 06, 07 and 09 are consequences of those results; they are printed once, as [lemma 19.3](#) and [theorem 19.4](#), and credited there. What is new relative to that report is only source 07’s classification of the class elementary self-embeddings of the full structure, which are not assumed surjective ([theorem 19.7](#)), and source 06’s two-lift classification of elementary embeddings between different cutoffs ([theorem 19.5](#)). No question named in that report is answered or refuted; in particular its question whether $\{\text{id}, c\}$ is a proper subgroup of its group Aut_L of field automorphisms commuting with $L(z) = \log|z|$ (equivalently, whether a nontrivial exponential automorphism of **No** exists) is untouched.

Product birthdays. No argument here uses a bound $b(xy) \leq b(x) \otimes b(y)$. That bound is Gonshor’s product-birthday conjecture; the collection’s [surreal/gonshor-product-birthdays](#) report proves it on a specified ring of ordinal-support series and states that the general conjecture is not settled. Sources 05, 06, 07 and 09 all say that they avoid it, and the size estimates below use only field closure of the epsilon-number cutoffs.

Sibling report. The report [foundations-and-computation/surreal-fields-across-universes](#), placed in the same batch, studies the external saturation of the *pure* ordered field $\mathbf{No}^M$ inside an outer universe, without a birthday symbol. Its setting meets this report’s outer-model part ([Part III](#)): the absoluteness of old arithmetic, the first new birthday, and the forcing examples. The two languages give opposite phenomena: the pure-field inclusion stays elementary but can lose saturation, while the birthday expansion loses elementarity ([remark 15.8](#)).

Other reports. The `foundations-and-computation/computable-surreals` report concerns countable effective subfields and is a different program; the undecidability remarks of [section 12](#) are standard. No report of the collection is contradicted by this one.

Repository statements of the sources, corrected. All five sources compared themselves with the collection at the pin `4f2645599121fa872c7104995e47f86f0382351f`, which is 32 commits before the placement commit `d4e71b7`.

- Sources 05 and 07 report that the catalogue at the pin described forty catalogued reports and four newly placed drafts, and that the formalization ledger counted forty-four main texts. This was true at the pin. Before this batch was placed the catalogue listed 44 research reports in five families.
- Sources 05 and 07 report that no matching reconstruction theorem was located in the material they inspected, and source 07 that the uniform bi-interpretation had not been located in earlier published work or in the repository. Source 05 read only the first 230 lines of the foundations report and source 07 its introduction; the Chen–Hamkins–Yang announcement is recorded in that report at the pin and now. The full-class statement is therefore credited to the announcement ([theorem 18.1](#)), and the local refinements of 05 and 07 carry the same priority caveat as those of 02, 06 and 09.
- Sources 05 and 06 cite line ranges and section structure of the surcomplex automorphisms report at the pin. That report has since been rewritten; its subsection “Simplicity restores the canonical surreal numbers” still exists, now in its section `saut:sec:exponential`, and the relative result is its `saut:thm:axis`.
- Source 06 reports that a GitHub search for “Hamkins” returned no indexed matches. That was a search-index observation; a direct search of the repository finds the announcement in the foundations report.
- Sources 02 and 07 report that the incoming-report directory contained only its README at the pin; that was true at the pin.

Source 02’s own source audit, kept byte-identical as `02-birthday-cutoffs-SOURCE_AUDIT.md`, contains no statement about the repository that is false; its closing description of the elementary-inclusion criterion as “necessary, not claimed sufficient” describes source 02 and is superseded inside this report by [theorem 9.3](#).

2 Foundations and the two ordinal languages

2.1 Ambient theory and class conventions

The proofs take place in ZFC, except in [section 13](#) (ZF) and in [section 18.2](#) (GBC, stated there). All cutoff structures are sets. The statements about the full surreal class are definable-class statements, understood one first-order formula at a time. Neither a set of all surreals nor an internally available truth predicate for the entire universe is assumed. Quotient interpretations use definable equivalence relations; the Mostowski collapse supplies set representatives when we compare them with H_κ . Global Choice is not used outside [section 18.2](#).

When a transitive model M is explicitly mentioned, its surreals are the sign sequences belonging to M , and cardinalities with superscript M are computed there. A statement proved internally in an arbitrary model of ZFC does not assert that the model’s well-founded relations are externally well founded. The outer-model comparison theorems explicitly use transitive models.

2.2 The carrier and the birthday function

A surreal is a function $s : \alpha \rightarrow \{-, +\}$ for an ordinal α . Comparison is lexicographic, with termination between $-$ and $+$. The all-plus sequence of length α is the embedded ordinal α . Throughout, $b(s)$ is the *surreal ordinal* corresponding to the length of s . Thus it is a function on the surreal carrier, not an extra ordinal sort. Set-theoretic lengths and their surreal embeddings are distinguished by context.

For an ordinal λ put $\mathbf{No}_{<\lambda} = \{x \in \mathbf{No} : b(x) < \lambda\}$, a set. For an epsilon number λ define

$$\mathcal{B}_\lambda = (\mathbf{No}_{<\lambda}; 0, 1, +, -, \cdot, <, b), \quad \mathcal{B} = (\mathbf{No}; 0, 1, +, -, \cdot, <, b).$$

The language includes no exponentiation, support predicate, membership predicate, or second-order quantifier.

Lemma 2.1 (Birthday order and birthday function). *On each $\mathcal{B}_\lambda$, and on $\mathcal{B}$, the birthday function and the birthday-precedence relation $x \prec_b y \iff b(x) < b(y)$ are definitionally interconvertible. So is the equal-birthday relation $E_b(x, y) \iff b(x) = b(y)$. Moreover, the embedded ordinals are exactly the fixed points of b .*

Proof. Given the function, precedence and equal birthday are immediate. Conversely, equality of birthdays is expressed by neither $x \prec_b y$ nor $y \prec_b x$. In the class of sign sequences of length α , the largest in numerical order is the all-plus sequence: any other sequence of that length has a first minus sign and is smaller. Therefore $b(x)$ is the unique largest member of x 's equal-birthday class,

$$y = b(x) \iff E_b(x, y) \wedge \forall z (E_b(x, z) \rightarrow z \leq y),$$

and this uses only E_b and the order. A sign sequence is fixed by this function exactly when it is all plus. Each cutoff contains the all-plus representative of each of its birthdays, so the definitions work in every $\mathcal{B}_\lambda$. $\square$

The interconvertibility with precedence is proved in sources 02, 06 and 09; the sufficiency of the coarser equal-birthday relation is source 07's observation. In particular the results below concern a definitional variant of the birthday-precedence language of the Chen–Hamkins–Yang announcement, not an essentially stronger coding apparatus. We abbreviate $\text{Ord}(a)$ by $b(a) = a$. Quantification over embedded ordinals in $\mathbf{No}_{<\lambda}$ is precisely quantification over $\alpha < \lambda$.

2.3 Natural arithmetic, not ordinary ordinal arithmetic

For ordinal arguments, surreal addition and multiplication are the Hessenberg operations

$$\alpha + \gamma = \alpha \oplus \gamma, \quad \alpha \gamma = \alpha \otimes \gamma.$$

They are strictly increasing in each positive argument where appropriate, and $\alpha + 1$ agrees with ordinary ordinal successor. We use ordinary ordinal exponentiation only in metamathematical bounds and in explicitly set-theoretic constructions. It is not silently identified with a field operation.

Lemma 2.2 (Closure and cardinal estimates). *Every uncountable cardinal is an epsilon number. For infinite ordinals α, γ , natural sums and products have cardinality at most $\max(\aleph_0, |\alpha|, |\gamma|)$. In particular, if κ is an uncountable cardinal and $\alpha, \gamma < \kappa$, their natural sum and product are below κ .*

Proof. In Cantor normal form, a natural sum joins finitely many exponent terms; a natural product uses finitely many natural sums of their exponents. The cardinality estimate follows from the corresponding estimate for ω^ξ : for infinite ξ , its cardinality is at most $\max(\aleph_0, |\xi|)$. One can see the latter by representing ordinals below ω^ξ by finite Cantor-normal-form lists with exponents below ξ and natural-number coefficients. These observations also cover finite arguments after adjoining $\aleph_0$ to the bound.

If κ is uncountable and initial, $\omega^\alpha < \kappa$ for every $\alpha < \kappa$. Continuity gives $\omega^\kappa \leq \kappa$. The opposite inequality follows from $\omega^\alpha \geq \alpha$ and taking suprema. Hence $\omega^\kappa = \kappa$. Neither step uses regularity of κ . $\square$

All five sources prove the first assertion; van den Dries and Ehrlich already note that every uncountable cardinal is an epsilon number [3], as source 06 records. For a general epsilon number, closure of natural arithmetic below λ also follows from the classical field-cutoff theorem. This simple closure is all the ordinal arithmetic needed by the relation codes.

2.4 Hereditary size rather than rank

Lemma 2.3 (Basic hereditary-size facts). *Let κ be an uncountable cardinal. A sign sequence of length α belongs to H_κ exactly when $\alpha < \kappa$. Every subset of an ordinal $\alpha < \kappa$ belongs to H_κ . Also $H_\kappa \subseteq V_\kappa$, and $\text{Ord} \cap H_\kappa = \kappa$.*

Proof. The transitive closure of the usual set-coded graph of a sign sequence has cardinality at most $\max(\aleph_0, |\alpha|)$, and it contains enough of the domain ordinal to have cardinality at least $|\alpha|$ when α is infinite. The same estimate applies to a subset of α , and $\text{TC}(\{\alpha\}) = \alpha + 1$.

If $|\text{TC}(\{x\})| \leq \mu$ for an infinite $\mu < \kappa$, the membership relation on this transitive closure is a well-founded relation of size at most μ . Its rank function takes values below μ^+ . This follows by well-founded induction, using regularity of the successor cardinal μ^+ : the supremum of at most μ ordinals below μ^+ remains below μ^+ . Thus $\text{rank}(x) < \mu^+ \leq \kappa$, and $x \in V_\kappa$. $\square$

The statement that every subset of α belongs to H_κ does *not* say that the collection $\mathcal{P}(\alpha)$ is an element of H_κ . This distinction drives the growth results. Three notions must not be interchanged: a family whose individual members are small, a family that is a small set, and a family that belongs to H_κ . For example, all subsets of ω belong to H_{ω_1} , but their collection does not (source 06).

3 Small certificates for surreal arithmetic

A reverse interpretation cannot simply say “perform the usual set-theoretic recursion inside H_κ .” At a singular cutoff, Replacement may fail in that structure. We instead prove that each individual arithmetic computation has a certificate of the same hereditary cardinal scale as its inputs. Two routes are given: source 09’s option-graph certificates (theorem 3.3) and the prefix-pair tables of sources 05, 06 and 07 (section 3.4).

3.1 Legal cuts can be checked using prefixes

For sets of sign sequences $L < R$, let $\{L \mid R\}$ denote their simplest separator. The classical simplicity theorem supplies this separator and the bound

$$\text{b}(\{L \mid R\}) \leq \sup\{\text{b}(a) + 1 : a \in L \cup R\}. \quad (3.1)$$

The supremum of the empty set is zero. These statements can equivalently be proved by the sign-by-sign cut construction [2].

Lemma 3.1 (Prefix test for a cut value). *A sign sequence z equals $\{L \mid R\}$ if and only if it separates L and R and no proper sign prefix of z separates them.*

Proof. The forward implication is immediate. For the converse, suppose y is a separator shorter than z . If y is a prefix of z , the stated test fails. If it is not, their longest common prefix lies numerically between them and is a proper prefix of z . The set of separators is convex, so this common prefix is another separator. Again the test fails. Thus z has minimum birthday among separators, and uniqueness of the simplest separator gives the result. $\square$

The check in this lemma quantifies only over the displayed option sets and the prefixes of z . In set coding it is a bounded check: one does not quantify over all sign sequences of earlier birthdays.

3.2 Well-founded option graphs of small size

A labeled option graph has a set D of vertices, left and right successor relations, and a rank label that strictly decreases along every option edge. A valuation of such a graph assigns a sign sequence to each vertex so that its value passes lemma 3.1 for the values at its options. When the graph is numeric, this valuation exists and is unique.

Lemma 3.2 (Small numeric graphs have small certificates). *Suppose μ is infinite and a numeric, well-founded option graph has at most μ vertices, with its structural data of hereditary size at most μ . Then it has a rank-labeled valuation certificate of hereditary size at most μ .*

Proof. The graph rank of every vertex is below μ^+ , by the successor-cardinal argument in lemma 2.3. Applying (3.1) by induction shows that a vertex value has birthday at most that vertex's graph rank. Thus each rank label and each sign value has hereditary size at most μ . There are at most μ vertices and at most $\mu^2 = \mu$ edges, so the union of all these data still has hereditary size at most μ . The existence construction here takes place in the ambient ZFC universe, not in a possibly deficient H_κ . $\square$

3.3 The computation graph for addition and multiplication

Take all sign prefixes of the inputs, including the inputs themselves. With

$$\mu = \max(\aleph_0, |\mathbf{b}(x)|, |\mathbf{b}(y)|),$$

this prefix family and its data have hereditary size at most μ . For addition, use finite signed words of prefix atoms $\mathbf{a}(a)$; an option moves one atom to one of its sign-prefix options, reversing left and right for a negative atom. The root word is $\mathbf{a}(x) + \mathbf{a}(y)$. This is the usual disjunctive-sum construction of the numeric game for $x + y$.

For multiplication, use finite signed words of pair atoms $\mathbf{p}(a, a')$, with a, a' prefixes of the respective inputs. An option of a positive pair atom is a signed three-term word

$$\mathbf{p}(a^\circ, a') + \mathbf{p}(a, a'^\circ) - \mathbf{p}(a^\circ, a'^\circ). \quad (3.2)$$

It is a left option when (a°, a'°) is left–left or right–right, and a right option for the two mixed choices. Negative atoms exchange the sides and negate the replacement word. An option of a word

replaces one atom by one of these words. The root is $\mathbf{p}(x, y)$. This spells out Conway's multiplication recursion without first constructing a large universe of games.

There are at most μ possible finite words. These graphs are well founded. For addition, use the natural sum, over atoms of the word, of $\omega^{b(a)}$. For multiplication put

$$\rho(a, a') = b(a) \oplus b(a')$$

and use the natural sum of $\omega^{\rho(a, a')}$ over the pair atoms. Every pair atom in (3.2) has strictly smaller ρ than the replaced atom. A finite sum of lower powers of ω is smaller than the replaced power, so each move strictly decreases the displayed measure. The standard numeric-game arithmetic theorem identifies the values of these graphs with $x + y$ and xy ; their options are exactly the classical recursive options, including the signs in the negative summands [1].

Theorem 3.3 (Hereditary-size arithmetic certificates). *There are fixed first-order set-theoretic formulas defining surreal addition, negation, and multiplication on sign sequences such that, for inputs of hereditary size at most an infinite μ , a correct computation has a certificate of hereditary size at most μ . Verification of a supplied certificate is absolute between transitive sets containing its data. Consequently, for every uncountable cardinal κ , these formulas define in H_κ exactly the external surreal operations on its sign sequences. No assumption $H_\kappa \models \text{Replacement}$ is required.*

Proof. For $+$ and multiplication, existentially quantify a set of word vertices containing the root and closed under the explicit option rule, its exact option relations, an ordinal rank labeling decreasing on edges, and a valuation by sign sequences passing the prefix cut test at every vertex. The node labels specify the input prefixes and the finite signed words; all checks of option generation quantify over those input prefix families and over entries of the finite words. Exactness requires both that each generated option is present and that every recorded edge is generated by the rule. The predicates for sign sequence, prefix, numerical comparison, finite word, and decreasing ordinal rank have bounded verifications on the supplied sets. In particular, a supplied prefix family is not accepted merely because its members are prefixes: for each ordinal $\xi \leq \text{dom}(s)$ it must contain the unique restriction $s \upharpoonright \xi$, and every recorded prefix must be one of these restrictions. The same completeness check is imposed on the prefix families of vertex values used by the cut test. These are bounded checks against $\text{dom}(s) + 1$ and the supplied families. A word length is a finite ordinal; this is checked by requiring every ordinal at most that length to be zero or a successor. Thus a missing prefix or a nonstandard-length word cannot evade an option requirement. The verifier is consequently a fixed bounded set-theoretic predicate, after including these domains and prefix families as certificate data. Their cardinality and hereditary-size bounds remain μ .

Any valid certificate is sound: external well-founded induction on its rank labels identifies each vertex value with the value of its generated numeric game. This also proves uniqueness of the root value. A certificate exists by the preceding graph construction and lemma 3.2, with hereditary size at most μ . Negation simply reverses all signs and has the same size estimate.

For inputs in H_κ , choose the infinite bound $\mu < \kappa$ given above. The external certificate belongs to H_κ ; conversely any certificate there is externally sound by bounded absoluteness. Hence the existential formulas have the same intended values. The uses of cardinal regularity were at μ^+ , not at κ . $\square$

Remark 3.4 (Scope of the certificate theorem). This is not a proof that every class-recursive surreal operation is absolute in every hereditary-size structure. It proves the specific finite-signature claim needed here. Inversion is definable from multiplication by $xy = 1$; its existence in the epsilon-number cutoffs, and real closedness, are supplied by the classical cutoff theorem. No normal-form birthday estimates or unproved sharp product-birthday conjecture enter the argument (section 1.5).

3.4 Second route: prefix-pair tables

Sources 05, 06 and 07 certify arithmetic by a single table indexed by pairs of input prefixes. Source 05 states it at regular cardinals; source 06 isolates the single-cardinal size bound that makes it work at every uncountable cardinal, singular ones included, and source 07 uses the same bound. We print this route with 06's bound.

For a sign sequence x write $\text{Pf}_x = \{x \upharpoonright \alpha : \alpha \leq \text{b}(x)\}$ for its prefix set, including x , and let L_x and R_x be its proper prefixes below and above x ; then $x = \{L_x \mid R_x\}$. Say that w is the *simplest separator* of two families when every left member is below w , every right member is above w , and no sign sequence of smaller length has the same property. For any candidate $w \in H_\kappa$, all sign sequences shorter than w also lie in H_κ (lemma 2.3); so this minimality test is absolute between H_κ and V whenever the families are in H_κ . Uniqueness of a least-birthday separator follows from signs: two distinct separators of the same least birthday would have a common prefix at their first difference strictly between them, still separating and of smaller birthday.

Lemma 3.5 (Hereditary size of arithmetic tables). *Let κ be an uncountable cardinal, let $x, y \in \mathbf{No}_{<\kappa}$ and set*

$$\mu = \max\{\aleph_0, |\text{b}(x)|, |\text{b}(y)|\} < \kappa.$$

The functions on $\text{Pf}_x \times \text{Pf}_y$ given by $T_{x,y}(u, v) = u + v$ and $U_{x,y}(u, v) = uv$ have graphs of hereditary size at most μ ; in particular they belong to H_κ .

Proof. There are at most μ pairs of prefixes. Every prefix belongs to $\mathbf{No}_{<\mu^+}$, which is a field by lemma 2.2 and the classical cutoff theorem. Therefore every value in either table has birthday below μ^+ and hereditary size at most μ . A set of at most μ objects, each of hereditary size at most μ , has hereditary size at most μ ; the finite layers needed for ordered pairs and triples do not change this.

This argument is carried out in the ambient ZFC universe to establish the existence of a particular witness in H_κ . It does not invoke Replacement inside H_κ , and it uses a *single* infinite cardinal μ , not a union of sizes approaching a singular κ . It also works when $\mu^+ = \kappa$. $\square$

Theorem 3.6 (Table definitions of the operations; sources 05, 06, 07). *Let $\text{Add}(x, y, z)$ say that there is a function T on $\text{Pf}_x \times \text{Pf}_y$ with $T(x, y) = z$ such that every entry obeys*

$$T(u, v) = \{T(u^L, v), T(u, v^L) \mid T(u^R, v), T(u, v^R)\} \quad (u^L \in L_u, v^L \in L_v, u^R \in R_u, v^R \in R_v),$$

*the braces meaning the simplest-separator predicate. Let $\text{Mult}(x, y, z)$ say that there is a function U on $\text{Pf}_x \times \text{Pf}_y$ with $U(x, y) = z$ such that $U(u, v)$ is the simplest separator of the left options $U(u', v) + U(u, v') - U(u', v')$ with $(u', v') \in L_u \times L_v \cup R_u \times R_v$ and the right options of the same form with $(u', v') \in L_u \times R_v \cup R_u \times L_v$, where each sum and difference is certified by **Add** and sign reversal. Then for every uncountable cardinal κ and all $x, y, z \in \mathbf{No}_{<\kappa}$,*

$$H_\kappa \models \text{Add}(x, y, z) \iff z = x + y, \quad H_\kappa \models \text{Mult}(x, y, z) \iff z = xy.$$

Proof. The actual tables are in H_κ by lemma 3.5 and satisfy the canonical Conway recursions, proving existence. Conversely, any accepted table is identified with the actual one by external well-founded induction on the pair of prefix lengths, ordered so that decreasing either coordinate decreases the pair: every option entry is earlier, the simplest-separator test is absolute, and the separator is unique. For multiplication, addition has already been identified. Each invocation of **Add** has its own hereditarily small witness; first-order satisfaction does not require that a single set collect the witnesses to all invocations. $\square$

The two routes certify the same operations; the table route uses fewer auxiliary objects, the option-graph route makes the verifier an explicitly bounded predicate. Either suffices for the reverse interpretation of [section 6](#).

Remark 3.7 (Regularity in source 05). Source 05 states its size bookkeeping and its table lemma for uncountable regular κ and declines to infer a singular-cardinal theorem. Its table argument uses only the field closure of $\mathbf{No}_{<\mu^+}$ and the single bound μ , exactly as in [lemma 3.5](#); regularity of κ is not used there. The singular case is sources 06 and 07's theorem, and the base's.

4 Definable sign access and a bounded pairing map

4.1 Recovering the sign tree from the two orders

For x, y, z let $\text{Between}(z; x, y)$ say that z lies in the closed numerical interval with endpoints x and y . Define

$$\text{Pre}(x, y) \iff \forall z ((\text{Between}(z; x, y) \wedge z \neq x) \longrightarrow b(x) < b(z)). \quad (4.1)$$

Lemma 4.1 (Definable prefix relation). *In every $\mathcal{B}_\lambda$, and in $\mathcal{B}$, (4.1) holds exactly when x is a sign prefix of y , allowing equality.*

Proof. If x is a prefix of y , every other point between them must continue the sign sequence of x ; a point diverging earlier would lie outside that interval. Its birthday is therefore larger.

If x is not a prefix of y and y is a proper prefix of x , the endpoint $z = y$ violates the formula. In the remaining case the longest common prefix of x, y lies strictly between them and has birthday smaller than that of x . This prefix belongs to the same cutoff. Finally, for $x = y$ the formula is vacuous, as required. $\square$

Remark 4.2 (Four prefix formulas). Each source defines prefix from $<$ and b by its own formula; all are correct. Source 02 requires $b(x) \leq b(y)$ and that every z with $b(z) < b(x)$ compares with x as with y ; source 06 adds the symmetric comparison. Sources 05 and 07 require $b(x) \leq b(y)$ and that no z strictly between x and y has $b(z) \leq b(x)$ (05) or $b(z) < b(x)$ (07). In the betweenness form the length conjunct cannot be dropped: for $x = 1$ and $y = 0$ no surreal of birthday at most 1 lies strictly between them, but 1 is not a prefix of 0 (source 05, which checks this case in its finite suite).

Define the bit predicate by

$$\begin{aligned} \text{Bit}(s, \alpha) \iff & \text{Ord}(\alpha) \wedge \alpha < b(s) \wedge \\ & \exists t (\text{Pre}(t, s) \wedge b(t) = \alpha \wedge t < s). \end{aligned} \quad (4.2)$$

The unique prefix of length α lies below s precisely when the sign of s at position α is plus. Thus

$$\{\alpha < \delta : \text{Bit}(s, \alpha)\}, \quad b(s) = \delta, \quad (4.3)$$

runs through *all* subsets of δ , once each, for every $\delta < \lambda$. This uses a first-order quantifier over numbers of one birthday, not a second-order quantifier in the language. The exact length matters: trailing minus signs are not discarded, so the code has an explicit domain.

4.2 Pairing without reconstructing ordinary ordinal multiplication

For a positive embedded ordinal θ , put

$$p_\theta(\xi, \zeta) = \theta\xi + \zeta, \quad \Lambda(\theta) = \theta^2 + \theta. \quad (4.4)$$

All operations in (4.4) are field operations, hence natural ordinal operations on these arguments.

Lemma 4.3 (Bounded pairing). *The map p_θ injects $\theta \times \theta$ into $\Lambda(\theta)$. For every epsilon number λ and every $0 < \theta < \lambda$, $\Lambda(\theta) < \lambda$. For infinite θ , $|\Lambda(\theta)| = |\theta|$.*

Proof. Suppose $\xi < \xi' < \theta$ and $\zeta, \zeta' < \theta$. Since $\xi + 1 \leq \xi'$ in ordinal order, positivity and distributivity give

$$\theta\xi + \zeta < \theta\xi + \theta = \theta(\xi + 1) \leq \theta\xi' \leq \theta\xi' + \zeta'.$$

For fixed ξ , cancellation gives injectivity in ζ . Also $\theta\xi + \zeta < \theta^2 + \theta$. Closure below λ follows from its being a field cutoff. The cardinal estimate is lemma 2.2, together with $\theta \leq \Lambda(\theta)$. $\square$

Source 02 uses the same pairing and bound (its B_θ). Consequently a sign sequence of birthday $\Lambda(\theta)$ codes a binary relation on θ . To make this code canonical for a given relation, require every plus position to be in the image of p_θ ; unused positions are minus. This last requirement is the first-order formula

$$\forall \rho < \Lambda(\theta) (\text{Bit}(r, \rho) \longrightarrow \exists \xi, \zeta < \theta [\rho = p_\theta(\xi, \zeta)]).$$

Here and below bounded ordinal quantifiers include the predicate Ord.

For a graph between two different ordinals $\delta, \epsilon < \lambda$, use $\theta = \delta + \epsilon + 1$, which is above both. Coding the graph in the $\delta \times \epsilon$ rectangle of this square again stays below λ . The construction requires no definable global pairing on the entire class of ordinals, and no reconstruction of ordinary ordinal arithmetic.

Remark 4.4 (Four pairings in the sources). The other sources pack relations differently; each variant works with the codes below after the obvious changes. Source 05 uses the rectangle map $p_a(i, j) = aj + i$ for $i < a$, $j < d$, with length $a(d + 1)$, which is the same idea with the roles of the coordinates exchanged. Source 06 uses the polynomial of lemma 4.5, and source 07 the radix pairing of lemma 4.6.

Lemma 4.5 (A one-polynomial pairing; source 06). *On ordinal surreals put $\varpi(\alpha, \gamma) = (\alpha + \gamma)^2 + \alpha$. Then ϖ is injective on pairs of ordinals, increasing in each coordinate, and maps $\kappa \times \kappa$ into κ for every uncountable cardinal κ .*

Proof. Put $\sigma = \alpha + \gamma$. Since $0 \leq \alpha \leq \sigma$, $\sigma^2 \leq \varpi(\alpha, \gamma) \leq \sigma^2 + \sigma$. If $\sigma < \sigma'$ are ordinals, then $\sigma + 1 \leq \sigma'$ and $\sigma^2 + \sigma < (\sigma + 1)^2 \leq \sigma'^2$. The intervals belonging to different sums are therefore disjoint. Equal values force equal sums, then equal first coordinates by subtraction, and finally equal second coordinates. Monotonicity follows from positivity and monotonicity of addition and squaring on nonnegative elements, and the bound is lemma 2.2. $\square$

Lemma 4.6 (Admissible radices; source 07). *Let $\text{Rad}(q)$ say that q is an ordinal, $q > 1$, and $a + a' < q$ for all ordinals $a, a' < q$. The ordinals satisfying Rad are exactly $q = \omega^\rho$ with $\rho > 0$; for each of them $p_q(a, a') = qa + a'$ injects $q \times q$ into q^2 . Both statements are absolute between $\mathcal{B}$ and every initial cutoff containing the ordinals in question. Below an epsilon number λ the admissible radices are cofinal and $q^2 < \lambda$ for each of them.*

Proof. For $q = \omega^\rho$ every ordinal below q has Cantor normal form with exponents below ρ , and natural addition introduces no new exponent. If $q > 1$ is not a single monomial with coefficient one, split its normal form into two nonzero proper parts (splitting a coefficient if needed) to get $a, a' < q$ with $a \oplus a' = q$; $q = 1$ is excluded. For $a < q$ the exponents of $q \otimes a$ are all at least ρ , those of $a' < q$ are below ρ , and cancellation of natural addition recovers a and a' ; moreover $qa + a' < q(a+1) \leq q^2$. For $\xi < \lambda$ the radix $\omega^{\xi+1}$ exceeds ξ and lies below λ , and $(\omega^\rho)^2 = \omega^{\rho \oplus \rho} < \lambda$ since $\lambda = \omega^\lambda$ is closed under natural addition below itself. All ordinals below an argument of an initial cutoff are present there, so the quantified test is unchanged. $\square$

5 Pointed relation codes and the exact interpreted universe

5.1 The definable domain of codes

A prospective code is a triple

$$C = (\delta, r, t), \quad 0 < \delta < \lambda, \quad t < \delta, \quad b(r) = \Lambda(\delta),$$

with δ, t embedded ordinals and with unused bits of r equal to minus. Write

$$E_C(i, j) \iff i, j < \delta \wedge \text{Bit}(r, p_\delta(i, j)).$$

The orientation is *child belongs to parent*: $i E_C j$ will mean that the object at node i is a member of the object at node j .

Require extensionality,

$$\forall i, j < \delta [\forall k < \delta (E_C(k, i) \leftrightarrow E_C(k, j)) \longrightarrow i = j], \quad (5.1)$$

and well-foundedness in the following form:

$$\begin{aligned} \forall u [b(u) = \delta \longrightarrow ((\exists j < \delta \text{Bit}(u, j)) \longrightarrow \\ \exists j < \delta [\text{Bit}(u, j) \wedge \forall i < \delta (\text{Bit}(u, i) \longrightarrow \neg E_C(i, j))])]. \end{aligned} \quad (5.2)$$

By (4.3), this is *actual* well-foundedness of the set relation, not just well-foundedness against a restricted collection of subsets.

Finally require generation by the root:

$$\begin{aligned} \forall u [b(u) = \delta \longrightarrow (\text{Bit}(u, t) \wedge \forall i, j < \delta [\text{Bit}(u, j) \wedge E_C(i, j) \longrightarrow \text{Bit}(u, i)] \\ \longrightarrow \forall i < \delta \text{Bit}(u, i))]. \end{aligned} \quad (5.3)$$

Thus no proper downward-closed subset of the graph contains the root. Let $\text{Code}(C)$ be the conjunction of these conditions. The triple is an ordinary finite-dimensional interpretation domain in the single-sorted structure $\mathcal{B}_\lambda$. Source 02 uses four-tuples with an extra domain word that selects a subset of θ as nodes, and source 07 four-tuples with a radix; in both, a “clean” or “no junk” condition plays the role of the unused-bit convention.

Lemma 5.1 (Rooted collapse). *Every valid code has a unique Mostowski collapse $\pi_C : \delta \longrightarrow T_C$ onto a transitive set. If $x_C = \pi_C(t)$, then*

$$T_C = \text{TC}(\{x_C\}), \quad |\delta| = |\text{TC}(\{x_C\})|.$$

Proof. Define by well-founded recursion

$$\pi_C(j) = \{\pi_C(i) : i E_C j\}.$$

Extensionality and well-founded induction show that π_C is injective: if a collision existed, choose one minimizing the maximum of the two node ranks; comparison of predecessor values gives equal predecessor sets, and extensionality identifies the nodes. Its image is transitive, and the displayed recursion gives an isomorphism. Uniqueness follows by well-founded induction as well; this is the usual Mostowski-collapse argument.

The nodes reachable from t by finitely many downward edges form a downward-closed subset containing t . By (5.3) this is all of δ . Under the collapse, these are exactly x_C and its iterated members. Thus the image is $\text{TC}(\{x_C\})$, giving the cardinal equality. $\square$

Root generation is not needed merely to encode arbitrary sets. It is useful here because it makes the hereditary-size cost exact rather than just an upper bound polluted by unrelated graph components.

5.2 Equality and membership are definable

For codes $C = (\delta, r, t)$ and $D = (\epsilon, s, u)$, define $C \sim D$ when a coded bijection $f : \delta \rightarrow \epsilon$ preserves E , sends t to u , and reflects E . Such a graph is expressed by a sign sequence in a square of width $\delta + \epsilon + 1$, as in lemma 4.3. The familiar function, bijection, and edge-preservation clauses use only bounded ordinal quantifiers and bits. Therefore $\sim$ is a single first-order relation.

For membership, let $C \in_{\mathcal{I}} D$ mean that there is a coded injection $f : \delta \rightarrow \epsilon$ such that

$$E_D(f(t), u), \tag{5.4}$$

$$E_C(i, j) \longleftrightarrow E_D(f(i), f(j)) \quad (i, j < \delta), \tag{5.5}$$

$$E_D(v, f(j)) \longrightarrow \exists i < \delta [E_C(i, j) \wedge f(i) = v] \quad (j < \delta, 0 \leq v < \epsilon). \tag{5.6}$$

In (5.6), the quantifier on v simply ranges over the embedded ordinals below ϵ . The clause says that the image contains *all* predecessors of its nodes, not merely some of them. Sources 06 and 07 use the same downward-closed image clause; source 02 instead defines membership through an isomorphism of the root with the cone of a predecessor, using a definable reachability relation.

Lemma 5.2 (Correctness of equality and membership). *For valid codes,*

$$C \sim D \iff x_C = x_D, \quad C \in_{\mathcal{I}} D \iff x_C \in x_D.$$

Proof. A root-preserving isomorphism gives equal collapsed roots by uniqueness of collapse. Conversely, if the roots agree, the two images are the same transitive closure by lemma 5.1, and $\pi_D^{-1}\pi_C$ gives the required isomorphism. Its graph has size at most $\max(|\delta|, |\epsilon|)$ and is codable within the same cutoff.

For membership, (5.5) and (5.6), followed by induction along E_C , imply $\pi_D(f(j)) = \pi_C(j)$ for all j . The root condition then gives $x_C \in x_D$. Conversely, if $x_C \in x_D$, the inclusion $\text{TC}(\{x_C\}) \subseteq \text{TC}(\{x_D\})$ transports through the collapse maps to an injection satisfying all three clauses. Its image is predecessor-closed, so (5.6) really does hold. The graph is again small enough to be coded below the cutoff. $\square$

The predecessor-full clause cannot be omitted; Appendix B gives the counterexample, which the finite suites of sources 06, 07 and 09 check.

Let $\mathcal{I}(\mathcal{B}_\lambda)$ be the quotient of valid codes by $\sim$, with the membership relation just defined. All quantifiers in its interpreted formulas are replaced by quantifiers over triples satisfying Code. No quotient class is assumed to be an element of the original structure.

5.3 Cardinal rounding

Theorem 5.3 (Exact collapse range). *For every epsilon number λ , the map $[C] \mapsto x_C$ is an isomorphism*

$$\mathcal{I}(\mathcal{B}_\lambda) \cong (H_{\kappa(\lambda)}, \in).$$

More precisely, the ordinals δ supporting rooted codes for a given set x are exactly those with

$$0 < \delta < \lambda, \quad |\delta| = |\text{TC}(\{x\})|. \quad (5.7)$$

Proof. Every code has $\delta < \lambda \leq \kappa(\lambda)$. Hence its collapse has hereditary size $|\delta| < \kappa(\lambda)$, by lemma 5.1. This proves one inclusion of the range.

Conversely, suppose $x \in H_{\kappa(\lambda)}$. Its nonempty transitive closure has some cardinal $\nu < \kappa(\lambda)$. By the definition of cardinal ceiling, $\nu < \lambda$: if λ is a cardinal this is immediate; otherwise $\nu \leq |\lambda| < \lambda$. Choose a bijection from ν onto $\text{TC}(\{x\})$, transfer membership to ν , and designate the node for x . The resulting graph is extensional, well founded, and generated by that node. It has a sign code of length $\Lambda(\nu) < \lambda$. Thus x occurs in the range.

For any other $\delta < \lambda$ equipotent to this transitive closure, the same construction works with a bijection from δ . Necessity in (5.7) was already proved by rooted collapse. Correctness of both relations is lemma 5.2. $\square$

Choice is used here only to well-order one transitive closure at a time; the interpretation keeps all valid codes and never selects one code per set. Source 02 proves this theorem one way, at every epsilon number, as a four-dimensional interpretation; source 07 proves the same range at every epsilon number (its cardinality plateaux, with $\theta(\lambda)$ in place of $\kappa(\lambda)$), also stating it forward only; sources 05 and 06 prove it at regular, respectively all uncountable, cardinals.

Remark 5.4 (Why the bound is sharp; source 02). No set of hereditary size $\kappa(\lambda)$ can be represented by these codes: its collapse would require at least $\kappa(\lambda)$ distinct nodes, while each domain is an ordinal below $\lambda \leq \kappa(\lambda)$. This obstruction is intrinsic to the stated interpretation. It is not a claim that no different interpretation in $\mathcal{B}_\lambda$ could represent a larger structure.

Example 5.5 (The empty set and a first nonempty code). The empty set uses one node, no edges, and root zero. With $\delta = 1$, the canonical relation string has birthday $\Lambda(1) = 2$ and both signs minus. Thus a code for the empty set is $(1, --, 0)$, not the surreal zero itself. For $\{\emptyset\}$ take $\delta = 2$, root 1, and the single edge $0E1$. The relation string has length 6 and its only plus is at $p_2(0, 1) = 1$. Numeric values and the sets they code must not be identified. A three-node example with two labelings is in Appendix C.

Corollary 5.6 (Countable cutoffs recover all countable sets; sources 02, 07, 09). *For every countable epsilon number λ , in particular ε_0 , $\mathcal{I}(\mathcal{B}_\lambda) \cong (H_{\omega_1}, \in)$.*

Proof. A countable epsilon number is strictly larger than ω and strictly below ω_1 , so its cardinal ceiling is ω_1 . Concretely, every hereditarily countable set has a rooted code on a finite ordinal or on ω , and $\Lambda(\omega) = \omega^2 + \omega < \varepsilon_0$. $\square$

The recovered sets include countable ordinals far larger than ε_0 . Those larger ordinals are recovered as *collapsed relations*, not as all-plus scalar sign sequences in the cutoff: every countable ordinal α has a membership graph labeled by natural numbers, whose relation word has birthday below $\omega^2 + \omega$. Labels need not preserve ordinal order; the collapse recovers that order from the edges. One should therefore not infer a low set-theoretic rank of the interpreted objects from a low birthday of the numerical codes (source 07).

6 Bi-interpretation and recovery of the cutoff

6.1 The ordinal and sign-graph bridges

For an actual embedded ordinal $a < \lambda$, there is a distinguished code $\text{OC}(a)$ with domain $a + 1$, root a , and edge relation iEj iff $i < j$. Its sign relation code is uniquely determined by the unused-bit convention. The formula defining $\text{OC}(a)$ is uniform, and its collapse is the von Neumann ordinal a . Because λ is a limit and a ring cutoff, both $a + 1$ and $\Lambda(a + 1)$ are below λ .

Definition 6.1 (Visible interpreted ordinals). For a set code C define

$$\text{VisOrd}(C) \iff \exists a [\text{Ord}(a) \wedge C \sim \text{OC}(a)].$$

Under the collapse isomorphism, these are exactly the ordinals below λ , not necessarily all ordinals of $H_{\kappa(\lambda)}$.

There is also a definable bridge $\text{SignGraph}(s, C)$ saying that C represents the set-coded sign function of the scalar s : its domain is $\text{OC}(b(s))$, its values are the interpreted ordinals 0 and 1, and at each actual $\alpha < b(s)$ it has value 1 exactly when $\text{Bit}(s, \alpha)$ holds. Application of a set-coded function and the Kuratowski ordered-pair construction are ordinary fixed formulas in the interpreted membership language. The ordinal bridge supplies their arguments. Thus SignGraph is a fixed first-order formula, not an external assignment operation.

For each s there is exactly one equivalence class of codes C with $\text{SignGraph}(s, C)$; conversely, every sign function whose length is below λ arises this way. Existence follows from the hereditary-size estimate for sign graphs and [theorem 5.3](#).

6.2 A first-order test for a cardinal cutoff

Define the *visible-ordinal collector* formula

$$\text{Coll}(C) \iff \text{Code}(C) \wedge \forall D [\text{Code}(D) \longrightarrow (D \in_{\mathcal{I}} C \leftrightarrow \text{VisOrd}(D))]. \quad (6.1)$$

It says that one interpreted set collects all the scalar ordinals. This is not the assertion that the scalar carrier itself is a set inside itself.

Theorem 6.2 (The cutoff is a set exactly in the noncardinal case; sources 02, 09). *For an epsilon number λ ,*

$$\mathcal{B}_\lambda \models \exists C \text{ Coll}(C) \iff \lambda \text{ is not a cardinal.}$$

When it exists, the collector is unique modulo $\sim$ and its collapse is exactly the ordinal λ .

Proof. The elements specified in (6.1) are exactly all ordinals $a < \lambda$. In the ambient set universe the unique set with precisely those elements is λ . Hence a collector exists precisely when $\lambda \in H_{\kappa(\lambda)}$, equivalently $|\lambda| < \kappa(\lambda)$, since $\text{TC}(\{\lambda\}) = \lambda + 1$. For a cardinal λ the right side fails; otherwise $\kappa(\lambda) = (|\lambda|)^+$ and it holds. Uniqueness follows from extensionality of the interpreted universe. $\square$

Source 02's sentence `CollectOrd` is this sentence written with its own codes. The theorem recovers the *ordinal value* of a noncardinal cutoff as an interpreted object. It is not a construction of a surreal number with birthday λ inside $\mathbf{No}_{<\lambda}$. This collector is the additional information needed to recover a noncardinal cutoff after interpreting its hereditary-size universe. Merely saying that $\mathcal{B}_\lambda$ interprets $H_{\kappa(\lambda)}$ would lose it.

6.3 The reverse interpretation

If $\lambda = \kappa$ is a cardinal, interpret surreals in H_κ as all set-coded functions from an ordinal into $\{0, 1\}$. By lemma 2.3, these are exactly the sign sequences of length below κ . Define the order by first disagreement, birthday by the all-one function on the domain ordinal, and arithmetic by the certificate formulas of theorem 3.3 (or the tables of theorem 3.6). Call this interpretation $\mathcal{J}$.

If λ is not a cardinal, work instead in the pointed structure $(H_{\kappa(\lambda)}, \in, \lambda)$. Use the same definitions, with the extra domain restriction that the sign length is below the named ordinal λ . The certificate verifier may use auxiliary sign sequences anywhere in $H_{\kappa(\lambda)}$; it is not required to carry out a large recursion inside the smaller interpreted sign domain. The root outputs are below λ by classical cutoff closure. This gives a uniform interpretation $\mathcal{J}_{\text{cut}}$ in the language with one named cutoff. Without the named cutoff, interpreting sign sequences in H_{ω_1} gives $\mathbf{No}_{<\omega_1}$, not $\mathbf{No}_{<\varepsilon_0}$; this is why source 07 stated its plateau theorem forward only.

Theorem 6.3 (Uniform local bi-interpretation). *The interpretations just constructed prove all assertions of theorem 1.1. The maps witnessing the two bi-interpretations are definable, not merely external bijections.*

Proof. Start on the set side. A code in the interpreted surreal structure gives an extensional well-founded relation on an ordinal. Its Mostowski-collapse root defines the map back to the original set. The graph of the collapse belongs to the same H_κ : its domain, range, and all values have hereditary size bounded by a fixed infinite cardinal below κ . Consequently this correspondence is defined by a first-order formula in H_κ , even when Replacement is not an axiom of that structure. The collapse equation is a bounded verification of the supplied function. Every set has a code, and equality and membership are respected by lemma 5.2. In the pointed case, the collector collapses to the named cutoff, so the correspondence preserves that constant as well.

Start on the surreal side. The formula $\text{SignGraph}(s, C)$ identifies s with the sign function representing it in the copy of the set universe. In the cardinal case all sign functions in the interpreted H_λ occur. In the noncardinal case, the collector defines λ , and the restricted reverse interpretation uses exactly lengths below that ordinal. Thus SignGraph is a definable bijection to the twice-interpreted surreal structure. It preserves numerical order and birthday directly from the sign rules, and it preserves arithmetic by the certificate theorem. This proves the required definable isomorphisms in both directions. $\square$

At cardinal cutoffs sources 06 and 07 give the same two round trips with their own codes (06 by a formula `Eval` for the collapse and a sign-function bridge; 07 by an ordinal-code comparison and the interpreted sign name $\langle b(x), A_x \rangle$), and source 05 gives them with graphs modulo bisimulation (section 7).

Corollary 6.4 (The first epsilon cutoff and hereditary countability). *The structure $\mathcal{B}_{\varepsilon_0}$ is parameter-free bi-interpretable with $(H_{\omega_1}, \in)$.*

Proof. The theorem first gives the pointed structure $(H_{\omega_1}, \in, \varepsilon_0)$. The ordinal ε_0 is parameter-free definable in H_{ω_1} as the least nonzero fixed point of ordinary $\alpha \mapsto \omega^\alpha$. For any countable arguments, the functions witnessing ordinal arithmetic and the relevant well-orders are countable sets, hence belong to H_{ω_1} . Their well-founded recursive verification is absolute. The least-fixed-point specification therefore picks out the actual ε_0 there. Naming it is a definitional expansion, which may be removed. $\square$

The same argument applies to any noncardinal epsilon number that is parameter-free definable, with its correct ordinal value, in the recovered hereditary-size universe. It is not an assertion that every ordinal can be named parameter-free.

Remark 6.5 (Questions of the sources answered here). Source 02 claimed only a one-way interpretation and warned that a two-way theorem “should specify whether the target carries a predicate for that segment or a constant for λ ”. [Theorem 6.3](#), source 09’s, is exactly such a theorem, with a constant; at ε_0 it is even unpointed. Source 06’s question on noncardinal cutoffs, source 07’s question on recovering a noncardinal cutoff, and the noncardinal half of source 05’s question on optimal cutoffs are answered for epsilon-number cutoffs by the same theorem; the singular-cardinal half of 05’s question is answered by sources 06 and 07. The questions stay open for cutoffs that are not epsilon numbers ([section 21](#)).

6.4 Coherence of the interpretation

Proposition 6.6 (Coherence under cutoff inclusions). *If $\lambda \leq \eta$ are epsilon numbers, the inclusion of scalar cutoffs induces, under $\mathcal{I}$, the actual inclusion*

$$H_{\kappa(\lambda)} \longrightarrow H_{\kappa(\eta)}.$$

A code keeps the same collapsed value, and all sign-graph bridges agree. For cardinal cutoffs the reverse interpretations also commute with the actual inclusions.

Proof. For a code already below λ , the larger structure has no additional sign sequences of birthday δ for any $\delta < \lambda$: both contain *all* sequences of each such length in the same ambient universe. Thus the subset tests in well-foundedness and root generation are unchanged. Equality and membership agree by their collapse characterizations; any needed witness graph has a code below λ as proved above. The collapse is defined from the same relation, so its value is unchanged. The sign-graph bridge refers to the same prefixes and bits. Finally the reverse arithmetic formulas compute the same external operations. $\square$

Sources 02 (at epsilon numbers, forward), 06 and 07 (at cardinal cutoffs, both directions) prove the same compatibility. If two noncardinal cutoffs have the same cardinal ceiling, the induced inclusion on recovered *unpointed* set universes is the identity. Their recovered distinguished cutoffs nevertheless differ. This is precisely why elementary inclusion will usually fail.

7 Second route: graphs modulo bisimulation

Source 05 codes sets by well-founded graphs that need not be extensional or generated by their root, and takes *bisimilarity* as equality. This avoids any extensionality or canonical-enumeration requirement: two different terminal nodes both represent $\emptyset$, and a disjoint component below no root changes nothing. We write its codes with the pairing of [lemma 4.3](#); source 05 uses its rectangle

map (remark 4.4), which changes nothing below. Source 05 proved the resulting bi-interpretation at uncountable regular cardinals; the argument below uses regularity nowhere (the certificates of section 3 are the only arithmetic input), and we state it at the cardinal cutoffs of theorem 1.1, where the base already proves the conclusion by the first route. The value of the second route is its different equality test.

A *graph code* is a triple $G = (\nu, e, t)$ with $\nu > 0$ and $t < \nu$ embedded ordinals and $b(e) = \Lambda(\nu)$; its relation is $E_G(i, j) \iff i, j < \nu \wedge \text{Bit}(e, p_\nu(i, j))$, again oriented from a member to a set. Let $\text{WF}(G)$ be the well-foundedness clause (5.2). No extensionality or root generation is required. The *decoration* of G is the unique function with

$$v_G(j) = \{v_G(i) : i E_G j\}, \quad \text{dec}(G) = v_G(t);$$

ordinary well-founded recursion suffices, and the Mostowski collapse is its extensional special case.

Proposition 7.1 (Exactly the hereditary sets; source 05). *For an uncountable cardinal κ , the decorations of graph codes in $\mathcal{B}_\kappa$ are exactly the members of H_κ .*

Proof. Every member of $\text{TC}(\{v_G(j)\})$ is a value at a node reachable by a finite predecessor chain, so it lies in the range of v_G , of cardinality at most $|\nu| < \kappa$. Conversely, pull membership on a transitive set containing $a \in H_\kappa$ back to an ordinal $\nu < \kappa$ along a bijection and root it at the label of a ; induction gives decoration a . $\square$

For graph codes $G = (\nu, e, t)$ and $G' = (\nu', e', t')$, a relation $Q \subseteq \nu \times \nu'$ is a *bisimulation* when every $(i, j) \in Q$ satisfies

$$\begin{aligned} i' E_G i &\implies \exists j' (j' E_{G'} j \wedge (i', j') \in Q), \\ j' E_{G'} j &\implies \exists i' (i' E_G i \wedge (i', j') \in Q). \end{aligned}$$

Code Q by one sign sequence q in a square of width $\nu + \nu' + 1$, and let $\text{Bis}(G, G', q)$ be the resulting first-order formula. Define

$$G \approx G' \iff \exists q [\text{Bis}(G, G', q) \wedge (t, t') \in Q_q], \quad G \in \approx G' \iff \exists j < \nu' [E_{G'}(j, t') \wedge G \approx G'[j]],$$

where $G'[j] = (\nu', e', j)$ is the same graph rerooted at j .

Theorem 7.2 (Equality is bisimilarity; source 05). *For well-founded graph codes, $G \approx G' \iff \text{dec}(G) = \text{dec}(G')$ and $G \in \approx G' \iff \text{dec}(G) \in \text{dec}(G')$. Consequently the graph codes modulo $\approx$, with $\in \approx$, form a parameter-free interpretation of $(H_\kappa, \in)$ in $\mathcal{B}_\kappa$.*

Proof. Suppose Q is a bisimulation relating the roots. By well-founded induction on i along E_G , every j with $(i, j) \in Q$ satisfies $v_G(i) = v_{G'}(j)$: a member $v_G(i')$ of $v_G(i)$ is matched by the forth clause with a predecessor j' of j and equals $v_{G'}(j')$ by induction, and the back clause gives the other inclusion. Conversely, if the root values agree, the set relation $Q = \{(i, j) : v_G(i) = v_{G'}(j)\}$ is a bisimulation relating the roots, because equal decorated sets supply the forth and back witnesses without selecting unique matches; it has a code in the same cutoff. Membership follows by applying this to the rerooted graph. Hence $\approx$ is an equivalence relation, a conclusion established semantically rather than presumed from the shape of the definition, and proposition 7.1 gives surjectivity. $\square$

Example 7.3 (Duplicates and irrelevant components). One terminal node represents $\emptyset$. A graph with two terminal nodes and a root above both represents $\{\emptyset\}$, as does a graph with only one terminal child. Their rooted graphs are not isomorphic, but their roots are bisimilar. Graph isomorphism is therefore *not* the right equality for these codes; source 05's finite suite exercises exactly this case.

The round trips of source 05 are the analogues of those in [theorem 6.3](#). The set-side map is the decoration relation “there is a function v on ν satisfying the recursion with $v(t) = a$ ”, whose witness has hereditary size at most $\max(\aleph_0, |\nu|)$. On the surreal side, for an ordinal α let $\text{OC}(\alpha)$ be the graph on $\alpha + 1$ with $iEj \iff i < j$ and root α , so that $G \approx \text{OC}(\alpha)$ iff $\text{dec}(G) = \alpha$; and for a surreal x let A_x be the interpreted set whose members are the classes $\text{OC}(\gamma)$ for the ordinals $\gamma < \text{b}(x)$ with $\text{Bit}(x, \gamma)$. The number x is then matched with the interpreted sign name $\langle \text{b}(x), A_x \rangle$, which externally decodes to $\langle \text{b}(x), \{\gamma : x(\gamma) = +\} \rangle$. Source 07 uses the same sign-name round trip with extensional codes.

Remark 7.4 (Internal versions and models of ZFC; sources 05, 07). All constructions above are theorems of ZFC and can be carried out inside any model $M \models \text{ZFC}$, with M ’s own ordinals, subsets and well-founded relations; for the full class $\mathbf{No}^M$ they recover M itself. For a nonstandard M this is an internal statement: graph well-foundedness is M ’s notion, not external well-foundedness. For set models M, N of ZFC, writing $\mathcal{B}^M$ and $\mathcal{B}^N$ for the internal birthday structures regarded externally, source 07 derives

$$M \equiv N \iff \mathcal{B}^M \equiv \mathcal{B}^N, \quad M \cong N \iff \mathcal{B}^M \cong \mathcal{B}^N,$$

by transporting sentences and isomorphisms through the uniform interpretations and their definable composites. These full-class and internal statements are instances of the announced Chen–Hamkins–Yang theorem ([theorem 18.1](#)), not results of sources 05 or 07.

8 Ring cutoffs and the finite boundary

The classical ring statement is weaker than the field statement: the infinite ring cutoffs $\mathbf{No}_{<\lambda}$ are exactly those with $\lambda = \omega^{\omega^\alpha}$ [[5](#), Theorems 1.1–1.2]. Source 02 observes that the forward interpretation needs only ring operations.

Proposition 8.1 (Ring version; source 02). *The interpretation $\mathcal{I}$ and its exact image remain valid for every infinite ordinal λ for which $\mathbf{No}_{<\lambda}$ is a subring of $\mathbf{No}$, when $\mathcal{B}_\lambda$ is regarded as a birthday-enriched ordered ring. Its image is $H_{\kappa(\lambda)}$, where now $\kappa(\lambda)$ is the least infinite cardinal $\geq \lambda$. In particular*

$$\mathcal{I}(\mathbf{No}_{<\omega}; +, -, \cdot, <, \text{b}) \cong (H_\omega, \in),$$

the hereditarily finite sets, and $\mathcal{I}(\mathcal{B}_{\omega^\omega}) \cong (H_{\omega_1}, \in)$.

Proof. Every formula of the interpretation uses only the ordered-ring operations and birthday. The bounds $\Lambda(\theta)$ and $\theta + \epsilon + 1$ remain in the cutoff by ring closure, and an infinite ring cutoff is a limit ordinal, so the ordinal codes also fit. All later coding and collapse arguments use only these facts and the cardinal-ceiling calculation. For $\lambda = \omega$ every domain is finite, all codes for its subsets and finite isomorphisms have finite length, and the represented transitive closures are exactly the finite ones. $\square$

Real closedness matters for the comparison with decidable field theories and for the surcomplex transfer, but it is not what enables the set coding. The decisive ingredients are complete sign layers, definable reading of those signs, and enough ordinal arithmetic to fit relation codes below the cutoff. Only the forward interpretation is claimed at ring cutoffs that are not epsilon numbers. At ω the cutoff is the dyadic ring, not a field, and it would be incorrect to reuse any real-closed-field argument there. Arbitrary ordinal cutoffs that are not ring cutoffs are not classified.

Part II

Consequences inside one universe

9 The exact elementary-inclusion spectrum

The elementary-inclusion result is stronger than the observation that a birthday expansion is not governed by real-closed-field quantifier elimination. It identifies exactly which initial inclusions are possible.

Lemma 9.1 (Noncardinal lower cutoffs never include elementarily; sources 02, 09). *If $\lambda < \eta$ are epsilon numbers and λ is not a cardinal, then $\mathcal{B}_\lambda \not\prec \mathcal{B}_\eta$.*

Proof. Choose a code C in $\mathcal{B}_\lambda$ for its visible-ordinal collector. Its collapse is λ , and it satisfies $\text{Coll}(C)$ there. By coherence it still collapses to λ in $\mathcal{B}_\eta$. But λ itself is now a scalar ordinal, since $\lambda < \eta$. Its interpreted ordinal code is visible, and is not a member of the set coded by C , because $\lambda \notin \lambda$. Thus $\text{Coll}(C)$ is false in the larger structure. This single formula, with the three entries of C as parameters, witnesses failure of elementarity. $\square$

Source 02’s witness is the weaker formula “the coded set contains every literal ordinal”, which is also true in $\mathcal{B}_\lambda$ and false in $\mathcal{B}_\eta$ for the same code. As source 02 observes, $\mathcal{B}_{\varepsilon_0}$ and $\mathcal{B}_{\varepsilon_1}$ interpret the same H_{ω_1} , yet their inclusion is not elementary: interpretation does not discard the additional first-order information of the original structure.

Lemma 9.2 (A cardinal lower cutoff cannot include elementarily into a noncardinal cutoff). *If λ is a cardinal and η is not, then $\mathcal{B}_\lambda \not\prec \mathcal{B}_\eta$.*

Proof. The parameter-free sentence $\exists C \text{ Coll}(C)$ is false in the first structure and true in the second, by [theorem 6.2](#). $\square$

Theorem 9.3 (Classification of elementary initial inclusions). *For epsilon numbers $\lambda < \eta$,*

$$\mathcal{B}_\lambda \prec \mathcal{B}_\eta \iff \lambda, \eta \text{ are cardinals and } H_\lambda \prec H_\eta.$$

Proof. The preceding two lemmas force both cutoffs to be cardinals; they are uncountable, since no countable cardinal is an epsilon number. For cardinal cutoffs, an elementary inclusion of the surreal structures induces an elementary inclusion of their interpreted set structures. By [proposition 6.6](#) this is the actual inclusion $H_\lambda \subseteq H_\eta$. Conversely, an elementary inclusion of those set structures induces an elementary inclusion of the uniformly interpreted sign structures. The reverse interpretation and its definable sign-graph isomorphism identify this with the original inclusion $\mathcal{B}_\lambda \subseteq \mathcal{B}_\eta$. $\square$

Remark 9.4 (Source 02’s open question, answered). Source 02 proved the forward implication (its necessary conditions) and asked whether they could be sharpened to a sufficient criterion, noting that any proof would have to control the reverse interpretation locally. Source 09’s [theorem 9.3](#) is that criterion: the reverse interpretation of [theorem 6.3](#) and the coherence of [proposition 6.6](#) give the converse. Between cardinal cutoffs the equivalence is also proved by source 05 (regular cardinals) and sources 06 and 07 (all uncountable cardinals).

Corollary 9.5 (Two kinds of elementary extension). *The ordered field inclusion $\mathbf{No}_{<\lambda} \subseteq \mathbf{No}_{<\eta}$ is elementary for all epsilon numbers $\lambda < \eta$. Adding birthday changes its elementary status according to [theorem 9.3](#). In particular,*

$$\mathbf{No}_{<\varepsilon_0} \prec \mathbf{No}_{<\omega_1}, \quad \mathcal{B}_{\varepsilon_0} \not\prec \mathcal{B}_{\omega_1}.$$

Proof. Both fields are real closed, so ordered-field quantifier elimination gives the first assertion. The birthday assertions are instances of the theorem. $\square$

Despite the last failure, both structures in that example are separately bi-interpretable with H_{ω_1} . There is no contradiction: a bi-interpretation does not make *every particular inclusion* elementary. The smaller structure defines an interpreted cutoff set; that formula changes its meaning under the displayed inclusion.

Example 9.6 (Two field-equivalent cutoffs separated by one sentence; sources 05, 07). Let χ say that every set admits an injection into ω (defined as the least infinite ordinal). Then $H_{\omega_1} \models \chi$ and $H_{\omega_2} \models \neg\chi$, since $\omega_1 \in H_{\omega_2}$. Hence the parameter-free sentence $\chi^{\mathcal{I}}$ gives $\mathcal{B}_{\omega_1} \not\equiv \mathcal{B}_{\omega_2}$, although their ordered-field reducts are elementarily equivalent and the field inclusion is elementary.

Corollary 9.7 (Transfer of elementary equivalence; source 06). *For uncountable cardinals κ, λ , $\mathcal{B}_\kappa \equiv \mathcal{B}_\lambda \iff H_\kappa \equiv H_\lambda$. The complete theories $\text{Th}(H_\kappa, \in)$ and $\text{Th}(\mathcal{B}_\kappa)$ are computably many-one reducible to each other by translations independent of κ (source 05, at regular cardinals).*

Proof. Translate sentences through the uniform interpretations in both directions; the translations are effective syntactic operations. $\square$

Remark 9.8 (Bi-interpretation versus elementary equivalence; source 05). Distinct membership structures can be bi-interpretable without being isomorphic or elementarily equivalent in their own language; Freire and Hamkins give relative-consistency instances involving H_{ω_1} and H_{ω_2} [10]. [Example 9.6](#) does not contradict this: the theorem here supplies one uniform interpretation between each cutoff and its own hereditary-size structure, and does not classify all interpretations between two different hereditary-size structures.

Corollary 9.9 (Comparison with the full class). *For an epsilon number λ , interpreted formula by formula,*

$$\mathcal{B}_\lambda \prec \mathcal{B} \iff \lambda \text{ is a cardinal and } H_\lambda \prec V.$$

Proof. The same collector obstruction excludes a noncardinal lower cutoff. For a cardinal lower cutoff, the uniform interpretation and reverse sign-graph interpretation extend to definable classes ([theorem 18.1](#)). The induced set map is $H_\lambda \subseteq V$, so the elementary equivalence follows for each formula. This uses no internal satisfaction class for V . $\square$

The condition $H_\lambda \prec H_\eta$ in [theorem 9.3](#) is not replaced here by a conjectural large-cardinal characterization. It is an exact reduction of one elementary-inclusion problem to another familiar one, together with the complete exclusion of every noncardinal initial cutoff. If both cutoffs are strongly inaccessible, $H_\rho = V_\rho$ and the condition reads $V_\lambda \prec V_\eta$ (source 05).

9.1 Elementary embeddings

Sources 05 (regular cardinals) and 06 (all uncountable cardinals) prove that the uniform bi-interpretation also transfers elementary embeddings that are not inclusions. We state the theorem at uncountable cardinals and give source 05’s explicit formula. Write $\text{name}(x) = \langle b(x), A_x \rangle$ with $A_x = \{\alpha < b(x) : x(\alpha) = +\}$ for the set-theoretic sign name of x .

Theorem 9.10 (Elementary-embedding correspondence; sources 05, 06). *For uncountable cardinals κ, λ there is a canonical bijection*

$$\text{Emb}_{\text{el}}((H_\kappa, \in), (H_\lambda, \in)) \longleftrightarrow \text{Emb}_{\text{el}}(\mathcal{B}_\kappa, \mathcal{B}_\lambda),$$

respecting identities and composition. For $j : H_\kappa \rightarrow H_\lambda$ the corresponding map e_j is given by

$$\text{name}(e_j(x)) = j(\text{name}(x)) = \langle j(b(x)), j(A_x) \rangle,$$

and for $e : \mathcal{B}_\kappa \rightarrow \mathcal{B}_\lambda$ the corresponding j_e is $j_e(x_C) = x_{e(C)}$, with e applied coordinatewise to codes. In particular $e_j(\alpha) = j(\alpha)$ on ordinals and $e_j(x)(j(\alpha)) = x(\alpha)$ for $\alpha < b(x)$.

Proof. Sign names, their order, birthday and arithmetic are uniformly defined in the membership language (section 6), so an elementary j maps sign names to sign names and preserves every translated formula; this proves elementarity of e_j . Since the code domain and $\sim$ are parameter-free, an elementary e maps codes to codes and equivalent codes to equivalent codes, so j_e is well defined, and applying $\mathcal{I}$ to each membership formula proves it elementary. The definable round-trip isomorphisms commute with elementary maps, because an elementary map preserves their defining formulas and their unique outputs modulo the equivalence relations; hence $e_{j_e} = e$ and $j_{e_j} = j$, and composition and identities are respected. The name of an ordinal α is $\langle \alpha, \alpha \rangle$, which gives $e_j(\alpha) = j(\alpha)$, and elementarity of Bit gives the last identity. $\square$

Remark 9.11 (Do not replace $j(A)$ by the pointwise image). For a nontrivial elementary set embedding, $j(A)$ need not equal $j^{\text{“}}A$. The formula uses the actual image *set* $j(A)$: the sign sequence $e_j(x)$ has domain $j(b(x))$ and may have coordinates outside $j^{\text{“}}b(x)$, which a pointwise transport of signs would leave undetermined.

Corollary 9.12 (Critical ordinals and the identity case; source 05). *Corresponding embeddings move exactly the same ordinals; when a least moved ordinal exists it is the same for both. An elementary embedding $\mathcal{B}_\kappa \rightarrow \mathcal{B}_\lambda$ fixing every ordinal below κ is the canonical inclusion.*

Proof. If e fixes all ordinals, then $b(e(x)) = e(b(x)) = b(x)$, and elementarity of the bit formula gives the same signs below that birthday. $\square$

The correspondence is a classification by reduction, not an existence theorem for such embeddings: it proves no new large-cardinal consistency statement and does not refute rank-into-rank hypotheses. A proper canonical inclusion moves no ordinal, so “proper embedding” must not be confused with “embedding having a critical point”. Every automorphism of $\mathcal{B}_\kappa$ is the identity (lemma 19.3); that statement overlaps the collection’s existing simplicity rigidity and is not counted as new.

10 Birthday growth and set-existence axioms

10.1 Birthday eternity

Definition 10.1 (The eternity scheme Et; source 09). For every formula $\varphi(a, x, \vec{p})$ in the birthday-field language, include the assertion that if γ is an embedded ordinal and

$$\forall a < \gamma \exists! x \varphi(a, x, \vec{p}),$$

then some embedded ordinal ϱ satisfies

$$\forall a < \gamma \forall x (\varphi(a, x, \vec{p}) \longrightarrow b(x) < \varrho).$$

The tuple $\vec{p}$ consists of arbitrary parameters from the structure.

This is a precisely specified version of birthday boundedness for definable ordinal-indexed functions. The analogy with the eternity principle in the Chen–Hamkins–Yang announcements is intentional; the cutoff classification below is proved directly, rather than imported from an unexpanded axiom scheme in the slides.

Theorem 10.2 (Eternity forces cardinality of an epsilon cutoff). *If λ is a noncardinal epsilon number, then $\mathcal{B}_\lambda \not\models \text{Et}$.*

Proof. Put $\mu = |\lambda|$, so $\mu < \lambda$ and $\kappa(\lambda) = \mu^+$. Choose a bijection $f : \mu \rightarrow \lambda$. Its graph has hereditary size at most μ , so it belongs to H_{μ^+} and has a code in $\mathcal{B}_\lambda$. Fix one such code as a parameter. Using the interpreted graph of f and the ordinal bridge, a single formula now defines a scalar map on the actual ordinal μ : at $a < \mu$ its value is the embedded ordinal $f(a) < \lambda$. Its birthdays exhaust λ , hence have no bound below λ . This violates the corresponding instance of Et. $\square$

At a singular cardinal κ , an external cofinal map into κ generally has hereditary size κ and need not belong to H_κ . Thus the same argument cannot simply be repeated with $\mu = \text{cf}(\kappa)$. Whether an internally definable cofinal map exists is the point of the next theorem.

10.2 The axioms that all hereditary-size universes already satisfy

Lemma 10.3 (The fixed background axioms). *For every uncountable cardinal κ , H_κ satisfies Extensionality, Empty Set, Pairing, Union, Infinity, Foundation, every Separation instance, and Choice. Moreover the actual transitive closure of each of its elements belongs to H_κ , and its cardinality is correctly computed there.*

Proof. Transitivity gives Extensionality and Foundation. Empty Set, Pairing, Union, and Infinity preserve hereditary size below κ . For any fixed formula and parameters, the externally defined subset of a set $a \in H_\kappa$ satisfying that formula in the set structure H_κ has hereditary size bounded by that of a (up to finitely many additional objects). It therefore belongs to H_κ , proving Separation, including formulas with unbounded quantifiers in that structure.

An ambient choice function for a family $a \in H_\kappa$ has its domain and chosen values inside the transitive closure of a , and its graph has the same infinite hereditary-cardinality bound. It belongs to H_κ . This proves Choice for set families, not a class-choice principle.

Writing $T = \text{TC}(\{x\})$, the transitive closure of T adds at most the set T to its already transitive data, so $T \in H_\kappa$. Its description as the least transitive set containing x as an element is correct

internally: the actual least such set is available there. A bijection between T and its least equipotent ordinal has hereditary size below κ , and any bijection asserted internally is externally a bijection. Minimality among ordinals is therefore also computed correctly. $\square$

10.3 Replacement is exactly eternity at cardinal cutoffs

Theorem 10.4 (Eternity–Replacement equivalence). *For every uncountable cardinal κ ,*

$$\mathcal{B}_\kappa \models \text{Et} \iff H_\kappa \models \text{every instance of Replacement.}$$

Proof. Assume first Replacement in H_κ . A definable scalar function on an ordinal $\gamma < \kappa$ corresponds under the bi-interpretation to a set-theoretically definable function assigning to $a < \gamma$ its unique sign graph. Replacement collects those sign graphs in one set of H_κ . The union of their domains, with one added successor for each length, is bounded below κ : otherwise the transitive closure of the collected set would contain at least κ ordinals. More explicitly,

$$\varrho = \sup\{\text{dom}(s) + 1 : s \text{ is an image sign graph}\} < \kappa$$

bounds their birthdays strictly. This proves the corresponding eternity instance; the empty-domain case is vacuous.

Conversely, assume eternity and consider a formula defining, in H_κ , a function $F : a \rightarrow H_\kappa$, where $a \in H_\kappa$ and the formula has fixed parameters. Choose a bijection $e : \delta \rightarrow a$ with $\delta < \kappa$. Its graph belongs to H_κ . For each $i < \delta$, put externally

$$\nu_i = |\text{TC}(\{F(e(i))\})|,$$

regarded as an initial ordinal. By [lemma 10.3](#), this is a uniquely defined ordinal of H_κ , with the correct external value. The definition of F , the parameter e , and the hereditary-cardinality formula translate into a definable scalar map $i \mapsto \nu_i$ in $\mathcal{B}_\kappa$. Here the ordinal bridge is available for every ordinal below κ . No definable choice of a code for $F(e(i))$ is required: its *hereditary cardinal* is unique.

Eternity gives a $\varrho < \kappa$ bounding all these ordinals. Ambient Replacement forms the actual image set $A = F[a]$, and its transitive closure has cardinality at most

$$\max(\aleph_0, |\delta|, |\varrho|) < \kappa.$$

Indeed it is contained, up to the image set itself, in the union of $|\delta|$ transitive closures, each of cardinality at most $\max(\aleph_0, |\varrho|)$. Thus $A \in H_\kappa$, proving the desired Replacement instance in that structure. This proof uses a fixed cardinal bound below κ ; it never assumes that an arbitrary union of fewer than κ small sets is small. $\square$

Corollary 10.5. *If κ is uncountable and regular, then $\mathcal{B}_\kappa \models \text{Et}$.*

Proof. Any externally given function on an ordinal below κ has fewer than κ birthdays in its image. Regularity bounds them below κ . This argument applies in particular to definable functions. $\square$

This corollary genuinely uses regularity; the equivalence of the theorem does not.

10.4 Power Set and packing a birthday fiber

Lemma 10.6 (Power Set in a hereditary-size universe; sources 02, 06, 09). *For every infinite cardinal κ ,*

$$H_\kappa \models \text{Power Set} \iff \kappa \text{ is strong limit } (2^\nu < \kappa \text{ for all cardinals } \nu < \kappa).$$

For an individual $\alpha < \kappa$, the actual power set $\mathcal{P}(\alpha)$ is an element of H_κ exactly when $2^{|\alpha|} < \kappa$. Regularity is not required.

Proof. Every subset of a set $a \in H_\kappa$ is already in H_κ , and the internal subset relation is the actual one by transitivity. So an internal power set of a , if it exists, is the full actual power set $\mathcal{P}(a)$. Its hereditary size is $2^{|\alpha|}$ up to harmless finite or countable bounds. This proves necessity (apply it to the cardinals $\nu < \kappa$) and the individual assertion. Conversely, for $x \in H_\kappa$ with κ uncountable, the transitive closure of $\mathcal{P}(x)$ has size at most $\max(\aleph_0, |\text{TC}(\{x\})|, 2^{|x|})$, which is below a strong-limit κ . If $\kappa = \omega$, power sets of hereditarily finite sets are hereditarily finite. $\square$

Theorem 10.7 (The translated Power Set sentence; sources 02, 06). *Let $\text{Pow}^\mathcal{I}$ be the $\mathcal{I}$ -translation of the Power Set axiom $\forall a \exists p \forall x (x \in p \leftrightarrow \forall y (y \in x \rightarrow y \in a))$; it is one fixed parameter-free sentence of the birthday-field language, not a scheme. For every epsilon number λ ,*

$$\mathcal{B}_\lambda \models \text{Pow}^\mathcal{I} \iff \lambda \text{ is a strong-limit cardinal.}$$

Among uncountable regular cutoffs it holds exactly at the strongly inaccessible ones.

Proof. By [theorem 5.3](#) and [lemma 10.6](#), $\text{Pow}^\mathcal{I}$ holds exactly when $\kappa(\lambda)$ is strong limit. If λ is not a cardinal, write $\nu = |\lambda|$; then $\kappa(\lambda) = \nu^+$ and Cantor's theorem gives $2^\nu \geq \nu^+$, so the condition fails. If λ is a cardinal, its ceiling is itself. Adding regularity yields strong inaccessibility. $\square$

Source 02 proves this at every epsilon number, source 06 at uncountable cardinals (its $\text{Pow}^\#$). It would be incorrect to remove the words “among uncountable regular cutoffs”: a singular strong-limit cardinal such as $\beth_\omega$ also satisfies $\text{Pow}^\mathcal{I}$, and [section 17](#) shows that the same enriched field can have a regular cutoff in one universe and a singular one in another. The sentence detects the strong-limit property, not regularity, Replacement, or full ZFC.

Definition 10.8 (The fiber-packing principle Pack; source 09). For every embedded ordinal α , there exist a positive embedded ordinal θ and a scalar r , with

$$\chi = \theta + \alpha + 1, \quad \text{b}(r) = \Lambda(\chi),$$

such that

$$\forall u [\text{b}(u) = \alpha \longrightarrow \exists \xi < \theta \forall \zeta < \alpha (\text{Bit}(u, \zeta) \leftrightarrow \text{Bit}(r, p_\chi(\xi, \zeta)))] \quad (10.1)$$

This is one first-order sentence, using the previously defined bit and pairing formulas; it is *not* the Power Set axiom. Its rows list every sign string of birthday α ; repetitions are allowed. Unused coordinates can be ignored. It does not use ordinary ordinal concatenation or a primitive power-set operation.

Theorem 10.9 (Exact packing threshold). *In $\mathcal{B}_\lambda$, the instance of (10.1) for a particular $\alpha < \lambda$ holds exactly when*

$$2^{|\alpha|} < \kappa(\lambda), \quad \text{equivalently, } 2^{|\alpha|} < \lambda,$$

where the cardinal on the left is compared as an ordinal in the second inequality. Consequently Pack holds exactly when λ is an uncountable strong-limit cardinal.

Proof. A packing table with $\theta < \lambda$ rows gives a surjection from θ onto $\mathcal{P}(\alpha)$, so $2^{|\alpha|} \leq |\theta| < \kappa(\lambda)$. Conversely, if that power-set cardinal is $\nu < \kappa(\lambda)$, then $\nu < \lambda$ by cardinal rounding. Choose an enumeration of the fiber by $\theta = \nu$ and encode its table in the square of width $\chi = \theta + \alpha + 1$. The square code stays below λ by closure, so the required r exists. The equivalence of the two inequalities follows because no cardinal lies in $[\lambda, \kappa(\lambda))$.

For a cardinal λ , validity for all $\alpha < \lambda$ is exactly the strong-limit property. If λ is not a cardinal, let $\mu = |\lambda| < \lambda$ and use $\alpha = \mu$. Cantor's theorem gives $2^\mu \geq \mu^+ = \kappa(\lambda)$, so this instance fails. $\square$

Notice that for a noncardinal cutoff λ , the recovered $H_{\kappa(\lambda)}$ and the scalar domain do not have the same ordinal height. Assertions about *all* interpreted ordinals and assertions about only scalar ordinals must be kept separate. The failure at $\alpha = |\lambda|$ is enough to settle the full packing principle.

Cutoff λ	Recovered universe	$\exists C$ Coll(C)	$\text{Pow}^{\mathcal{I}}$, Pack
Countable epsilon number	H_{ω_1}	true	false
Noncardinal, $ \lambda = \nu$	H_{ν^+}	true	false
Uncountable successor cardinal κ	H_κ	false	false
Singular strong-limit cardinal κ	H_κ	false	true
Strongly inaccessible κ	$H_\kappa = V_\kappa$	false	true

The table is source 02's, with the packing column added from source 09.

11 Worldly cutoffs and separation examples

11.1 The exact conjunction

Call an uncountable cardinal κ *worldly* here when $V_\kappa \models \text{ZFC}$. This definition does not impose regularity. We need the following familiar set-theoretic fact, for which a proof is included to fix the precise meaning.

Lemma 11.1 (When hereditary size becomes cumulative rank). *For an uncountable cardinal κ ,*

$$H_\kappa \models \text{ZFC} \iff H_\kappa = V_\kappa \text{ and } V_\kappa \models \text{ZFC}.$$

Moreover, $V_\kappa \models \text{ZFC}$ by itself implies $H_\kappa = V_\kappa$.

Proof. Suppose $H_\kappa \models \text{ZFC}$. Its internal cumulative hierarchy exists to every ordinal $\alpha < \kappa$. Every subset of each of its sets belongs to H_κ , so its internal power sets are actual power sets. Induction on α shows that its internal V_α is the actual V_α ; the limit step is ordinary union over the actual smaller ordinals. Thus every V_α for $\alpha < \kappa$ belongs to H_κ , and transitivity gives $V_\kappa \subseteq H_\kappa$. The reverse inclusion is [lemma 2.3](#).

Conversely, assume $V_\kappa \models \text{ZFC}$. For every $\alpha < \kappa$, that model can well-order V_α and supply a bijection with some ordinal $\delta < \kappa$. These are actual sets and actual bijections, because the model is transitive. Hence $|V_\alpha| < \kappa$. The transitive closure of any $x \in V_\kappa$ is contained in some such V_α , and therefore has size below κ . Again $H_\kappa = V_\kappa$ follows. The remaining implication is immediate from equality. $\square$

Theorem 11.2 (Worldly-cutoff characterization). *For an epsilon number λ ,*

$$\begin{aligned} \mathcal{B}_\lambda \models \text{Et} + \text{Pack} &\iff \lambda \text{ is a cardinal and } H_\lambda \models \text{ZFC} \\ &\iff \lambda \text{ is a worldly cardinal.} \end{aligned}$$

For regular uncountable λ , this holds exactly when λ is strongly inaccessible.

Proof. Either growth principle excludes noncardinal epsilon cutoffs. At cardinal cutoffs, [theorem 10.4](#) and [lemmas 10.3](#) and [10.6](#) identify the two missing ZFC requirements as Replacement and Power Set. The equivalence with the rank formulation is [lemma 11.1](#). For regular uncountable cardinals, eternity is automatic and packing is exactly strong limit, giving the usual inaccessible condition. $\square$

This proves [theorem 1.3](#). It does not identify the two formulas with every formulation of a theory called SA. It is an exact theorem about the explicit scheme and sentence defined here.

11.2 The two growth principles do different work

Example 11.3 (Eternity without fiber packing). At $\kappa = \omega_1$, regularity gives eternity. The birthday- ω fiber has $2^{\aleph_0} \geq \aleph_1$ members, so no ordinal below ω_1 can index a packing of it. Hence

$$\mathcal{B}_{\omega_1} \models \text{Et}, \quad \mathcal{B}_{\omega_1} \not\models \text{Pack}.$$

In the decoded structure, every individual real is present, but the set of all reals is not an element of H_{ω_1} .

Example 11.4 (Fiber packing without eternity). Let $\kappa = \beth_\omega = \sup_{n < \omega} \beth_n$, with $\beth_0 = \aleph_0$ and $\beth_{n+1} = 2^{\beth_n}$. This is a strong-limit cardinal: for every $\mu < \kappa$ some $\beth_n$ is at least μ , and then $2^\mu \leq \beth_{n+1} < \kappa$. Thus packing holds.

In H_κ , each finite initial segment of the beth recursion is present and correctly computed. The necessary finite sequence of power sets, cardinals, and bijections has hereditary size below κ , and Power Set is correct there by [lemma 10.6](#). A single formula therefore defines $n \mapsto \beth_n$ on the actual ω . Transporting this formula through the ordinal bridge gives a definable scalar function whose birthdays are cofinal in κ . Consequently

$$\mathcal{B}_{\beth_\omega} \models \text{Pack}, \quad \mathcal{B}_{\beth_\omega} \not\models \text{Et}.$$

The infinite sequence of all these ordinals is precisely what cannot be collected as an element of H_κ .

Source 06 isolates the same phenomenon as one explicit sentence, and uses it to separate $\beth_\omega$ from its successor. Let $\vartheta(n, \gamma)$ be the membership formula saying that a finite sequence of length $n + 1$ begins with ω , ends with γ , and at each step takes the least ordinal equinumerous with the power set of the previous entry; and let

$$\text{Rep}_\beth : [\forall n \in \omega \exists! \gamma \vartheta(n, \gamma)] \longrightarrow \exists y \forall \gamma [\gamma \in y \leftrightarrow \exists n \in \omega \vartheta(n, \gamma)],$$

a single Replacement instance.

Theorem 11.5 (A singular Replacement boundary; source 06). *With $\kappa = \beth_\omega$, $H_\kappa \not\models \text{Rep}_\beth$ and $H_{\kappa^+} \models \text{Rep}_\beth$, the antecedent being true in both. Consequently*

$$\mathcal{B}_{\beth_\omega} \not\prec \mathcal{B}_{\beth_\omega^+},$$

although $\mathcal{B}_{\beth_\omega} \models \text{Pow}^\mathcal{I}$ and the ordered-field inclusion is elementary.

Proof. Every finite beth computation, with the power sets and bijections at its finitely many stages, has hereditary size below κ and belongs to both structures, where it is correctly computed; so $\vartheta(n, \gamma)$ picks out the actual $\beth_n$ in both. A set collecting all $\beth_n$ has every ordinal below κ in its transitive closure, so it is not in H_κ ; the actual set $\{\beth_n : n < \omega\}$ has hereditary size κ and lies in $H_{\kappa+}$. Transfer by [theorem 9.3](#) (or the translation $\text{Rep}_{\beth}^{\mathcal{I}}$) gives the failure of elementarity, and [theorem 10.7](#) the Power Set assertion. $\square$

The theorem gives one singular example with failed Replacement. It does *not* show that every singular κ fails Replacement in H_κ : an externally cofinal sequence can fail to be definable over H_κ or fail to be an element of it. Nor does internal ZFC imply external inaccessibility; [corollary 17.2](#) gives a sharp example.

Example 11.6 (A noncardinal field cutoff). Both principles fail at ε_0 . Packing fails at the ordinal ω . Eternity fails by [theorem 10.2](#); one may also use the definable finite power-tower sequence cofinal in ε_0 , with ordinary ordinal exponentiation computed in the decoded H_{ω_1} . Nevertheless the field is real closed and its birthday expansion is bi-interpretable with hereditary countability.

These are actual structures exhibiting independence of the two displayed growth requirements within this family. They are not an assertion of independence from an unspecified list of all the other axioms of SA.

11.3 Singular cutoffs satisfying both principles

Proposition 11.7 (A conditional source of singular worldly cutoffs). *If Ω is strongly inaccessible, there is a cardinal $\kappa < \Omega$ of cofinality ω such that*

$$V_\kappa \prec V_\Omega.$$

In particular, κ is worldly and $\mathcal{B}_\kappa$ satisfies both growth principles.

Proof. Enumerate all existential formulas in the language of set theory. Construct strictly increasing infinite cardinals $\alpha_n < \Omega$. Given α_n , consider the first n formulas and all their finite parameter tuples from V_{α_n} . Whenever V_Ω has a witness, choose one. There are fewer than Ω such requirements, since $|V_{\alpha_n}| < \Omega$. Regularity bounds their witness ranks below Ω . Choose a cardinal $\alpha_{n+1} < \Omega$ above that bound and above α_n .

Set $\kappa = \sup_n \alpha_n$. Regularity of Ω gives $\kappa < \Omega$; it is a cardinal of cofinality ω . Every finite tuple in V_κ appears in one stage, and every existential formula is treated at all sufficiently late stages. Thus every existential statement true in V_Ω with parameters in V_κ has a witness in V_κ . The Tarski–Vaught criterion gives elementarity. Since V_Ω satisfies ZFC, so does V_κ . Apply [theorem 11.2](#). $\square$

The reflection construction is standard set-theoretic reasoning, included as an illustration rather than as a novelty claim, and it needs the stated inaccessible. It demonstrates why replacing “worldly” by “inaccessible” in [theorem 11.2](#) would be wrong. No inaccessible cardinal is assumed in the unconditional cutoff theorems.

12 Model-theoretic consequences

12.1 Full second-order arithmetic and undecidability

Proposition 12.1 (The standard natural numbers and all their subsets; sources 06, 07, 09). *In $\mathcal{B}$ and in every $\mathcal{B}_\lambda$ (λ an epsilon number) there is a parameter-free interpretation of the two-sorted structure $(\mathbb{N}, \mathcal{P}(\mathbb{N}); 0, 1, +, \cdot, <, \in)$ with its full actual subset sort. The number sort consists of the ordinals below the least nonzero limit ordinal w (externally $w = \omega$), the subset sort of the s with $b(s) = w$, and membership is $\text{Bit}(s, n)$.*

Proof. “ q is a nonzero limit ordinal” is $\text{Ord}(q) \wedge q > 0 \wedge \forall a < q \exists d (a < d < q)$ with ordinal guards, and w is its least solution. The ordinals below w are the standard finite ordinals, and natural arithmetic on them is ordinary arithmetic. The fiber $b(s) = \omega$ is in bijection with all subsets of ω by (4.3); no restriction to definable subsets is imposed. $\square$

When working inside a model M , the subset sort is $\mathcal{P}(\omega)^M$, as it must be. Sources 06 and 07 state the proposition, and the next two results, at uncountable cardinals and source 09 at ε_0 ; the proofs use only $\omega < \lambda$, which holds at every epsilon number. This is an ordinary first-order interpretation of a two-sorted structure, not a replacement of first-order by second-order syntax. It does not make set theory definable in the pure ordered-field reduct, where the birthday structure is absent.

Corollary 12.2 (Undecidability; sources 05, 06, 07, 09). *True first-order arithmetic reduces effectively to the complete theory of every $\mathcal{B}_\lambda$, and formula by formula to $\mathcal{B}$. Consequently these complete theories are neither decidable nor computably enumerable.*

Proof. Relativize arithmetic quantifiers to the number sort. If the target theory were computably enumerable, the inverse image would enumerate true arithmetic; a complete computably enumerable theory is decidable, and true arithmetic is not. $\square$

These are logical-complexity consequences, not new algorithms. A code can contain an arbitrary noncomputable subset of an ordinal, and first-order definability in the birthday expansion must not be confused with decidability from a finite symbolic representation.

12.2 Independence and an omitted type

Proposition 12.3 (Independence property and a countable omitted type; source 07). *The formula $\text{Bit}(s, n)$ has the independence property in every $\mathcal{B}_\lambda$. The parameter-free partial type $\{x < w \wedge \text{Ord}(x)\} \cup \{x > n : n \in \mathbb{N}\}$, with each standard n named by its definition, is finitely satisfiable but omitted.*

Proof. For distinct finite ordinals $n_1, \dots, n_k$ and any $S \subseteq \{1, \dots, k\}$, a length- ω sign code positive exactly at the n_j with $j \in S$ realizes the pattern. Finite parts of the type are realized by large finite ordinals; a realization of the whole type would be a standard natural number above all standard natural numbers. $\square$

Remark 12.4 (Saturation of the pure field, and the sibling report). For regular uncountable κ , the pure real closed field $\mathbf{No}_{<\kappa}$ realizes every one-variable type over fewer than κ parameters: by quantifier elimination a nonalgebraic type over the real closed subfield generated by the parameters

is a cut, regularity bounds the birthdays of its fewer than κ endpoints, and (3.1) supplies a separator. Source 07 records this as the usual saturation property of the pure field, not a new result. Naming birthday nevertheless produces the omitted type of [proposition 12.3](#). External saturation of the pure field $\mathbf{No}^M$ inside an outer universe is the subject of the sibling report [foundations-and-computation/surreal-fields-across-universes](#); this report does not study it.

12.3 Birthday is not visible in the field reduct

Proposition 12.5 (Nondefinability of birthday; sources 05, 07). *On $\mathbf{No}_{<\lambda}$, for λ an epsilon number, neither birthday nor prefix simplicity is definable, even with parameters, in the ordered-field reduct, nor in any o-minimal expansion of it. The same holds formula by formula on $\mathbf{No}$.*

Proof. If birthday were definable, its fixed-point set, the embedded ordinals, would be definable. It is infinite and contains no interval: between ordinals $\alpha < \gamma$ lies $\alpha + \frac{1}{2}$, which is not an ordinal. An o-minimal structure (in particular a real closed field, by quantifier elimination) has only finite unions of points and intervals as definable subsets of the line. If prefix simplicity were definable, the ordinals would be defined by “every proper prefix of x lies below x ”, with the same contradiction. $\square$

Source 05 states the proposition at regular cardinals and source 07 at uncountable cardinals; the proof uses only real closedness (o-minimality). The collection’s foundations report notes that the field language has no birthday symbol; this proposition is the corresponding standard nondefinability proof, new to the collection but not a novelty claim.

12.4 How little arithmetic is needed initially

Let $G_\varpi(\alpha, \gamma, \rho)$ be the ternary relation on $\mathbf{No}_{<\kappa}$ saying that α, γ, ρ are ordinals and $\rho = \varpi(\alpha, \gamma)$, with ϖ the polynomial of [lemma 4.5](#).

Corollary 12.6 (Reconstruction from ordinal pairing; source 06). *For every uncountable cardinal κ , all operations and constants of $\mathcal{B}_\kappa$ are uniformly definable in $(\mathbf{No}_{<\kappa}; <, b, G_\varpi)$. Hence the two structures are definitionally equivalent.*

Proof. Run the coding of [section 5](#) with relation words of length $\varpi(\delta, \delta) + 1$ whose plus positions are the $\varpi(u, v)$, as source 06 does; ordinal successor is definable from the ordinal order. The code formulas then use field arithmetic only to evaluate ϖ on ordinals, which G_ϖ supplies, so this reduct still interprets H_κ . Translate the arithmetic definitions of [section 3](#) through that interpretation and use the surreal-side round trip to put their graphs on the original carrier. The converse is the polynomial formula for G_ϖ . $\square$

No minimal-language claim is made: the corollary does not remove birthday data or show that arithmetic is definable from order and prefix simplicity alone. It is partial progress on the minimal-signature questions of sources 02, 05 and 07, which stay open ([section 21](#)).

13 The exact Choice boundary of ordinal graph coding

This section, source 06's, works in ZF . It considers the full class of sign sequences and the graph interpretation of [section 5](#) without a cutoff. All its defining formulas remain first-order formulas about individual sets or surreals, and class assertions are read formula by formula.

Definition 13.1. Let $HWO = \{x : TC(\{x\}) \text{ admits a well-ordering}\}$. One well-ordering of the whole transitive closure is required, with no definability condition such as the one in HOD .

Theorem 13.2 (Choice-sensitive range; source 06). *Over ZF , the pointed ordinal-graph interpretation has quotient structure isomorphic to $(HWO, \in)$. Its external collapse map is surjective onto V if and only if the Axiom of Choice holds.*

Proof. The Mostowski collapse of a valid code (δ, r, t) is injective with domain the ordinal δ , and its range is $TC(\{x_C\})$ by root generation; transporting the ordinal order well-orders that closure. Conversely, a well-order of $TC(\{x\})$ gives a bijection from an ordinal, along which membership pulls back to a valid code for x ; no simultaneous choice for a family is used. Equality and membership remain correct in ZF : all subsets of a fixed ordinal exist and have characteristic words, so the well-foundedness test still quantifies over every subset, and Mostowski collapse is a theorem of ZF . If Choice holds every transitive closure is well-orderable. Conversely, surjectivity well-orders every $TC(\{x\})$ and hence every $x \subseteq TC(\{x\})$, and the well-ordering principle implies Choice. $\square$

Proposition 13.3 (Internal Choice is a different assertion; source 06). *In ZF , HWO is transitive, contains every ordinal and every subset of an ordinal, and internally satisfies the statement that every set can be well-ordered. If $\mathcal{P}(\omega)$ is not well-orderable in the ambient universe, $(HWO, \in)$ fails the Power Set axiom at ω .*

Proof. If $y \in x \in HWO$ then $TC(\{y\}) \subseteq TC(\{x\})$ is well-orderable. For $a \subseteq \alpha$, $TC(\{a\}) \subseteq \{a\} \cup \alpha$. For $x \in HWO$ the restriction to x of a well-order of $TC(\{x\})$, and every subset of x , lie in HWO , since finite products and finite subsets of a well-orderable set are well-orderable in ZF ; so the internal well-order has a witness. Every subset of ω is in HWO , so an internal power set of ω would be the actual $\mathcal{P}(\omega)$, which is then well-orderable, contrary to the hypothesis. $\square$

The surjectivity statement of [theorem 13.2](#) is an external comparison of ranges, not the value of the internal Choice sentence. The theorem is about the specified ordinal-graph construction; no claim is made that every possible interpretation of sets in surreals has this boundary. Global Choice is unnecessary here and for the local ZFC bi-interpretations: existence of a code is proved object by object and the quotient identifies all codes.

Remark 13.4 (Source 09's fourth target, answered). Source 09 listed “the choice-free analogue” among its targets and observed that ordinal-indexed relations naturally recover the sets with well-orderable transitive closures. [Theorem 13.2](#), source 06's, is that analogue for the full class; source 07 records the same range in a remark without a separate proof. Whether a richer coding signature can escape the boundary, and a choice-free version of the *cutoff* theorems, remain open.

Part III

Outer models

Throughout this part, $M \subseteq N$ are transitive models of ZFC with the same ordinals. They may be set models viewed from an ambient metatheory, or class models with formulas interpreted one at a time. Superscripts indicate the model in which a construction is performed, and all references to M -cardinality are internal to M . The intended examples are set-forcing extensions, but no result below assumes forcing unless it says so.

14 Old arithmetic and cross-model coherence

Lemma 14.1 (An old simplest separator stays simplest; sources 02, 05, 07). *If $L, R \in M$ are separated sets of old sign sequences and $w \in M$ is their simplest separator computed in M , then w is their simplest separator in N .*

Proof. Comparisons of old sign sequences are absolute, so w still separates. Suppose $z \in N$ separates and is shorter. If z is a prefix of w it is a restriction of w to an ordinal, hence old, contradicting minimality in M . Otherwise z and w first disagree before z ends; their common prefix p is a restriction of w , hence old, lies strictly between z and w , and is a shorter separator by convexity of the set of separators, with all its comparisons with L, R holding in M . This again contradicts minimality. $\square$

Source 02 gives a constructive variant: build the separator sign by sign, appending $+$ if some $\ell \in L$ is at least the current word and $-$ otherwise, and taking unions at limits; every test involves only the given sets and the previous prefix, so the stopping place and every sign agree in M and N .

Proposition 14.2 (Absolute old field structure; sources 02, 05, 07). *The inclusion of old sign sequences $\mathbf{No}^M \subseteq \mathbf{No}^N$ preserves and reflects order and preserves birthday, constants, negation, addition, multiplication and inverses. Its ordered-field reduct is an elementary inclusion, and so is the pure-field inclusion $\mathbf{No}^M[i] \subseteq \mathbf{No}^N[i]$. The same holds for $\mathbf{No}_{<\lambda}^M \subseteq \mathbf{No}_{<\lambda}^N$ at every common epsilon number λ .*

Proof. Order, length, constants and sign reversal are absolute. For addition induct on pairs of old prefixes: all prefixes of an old number are old, the options are identified by induction, and lemma 14.1 identifies their simplest separator. Multiplication follows by the same induction using addition, and inverses by uniqueness. Both real fields are real closed and both complex fields algebraically closed of characteristic zero, so quantifier elimination makes the inclusions elementary [14, 15]. For proper classes the equivalences are applied one formula at a time. $\square$

This is the usual absoluteness of the canonical construction, not an assertion about arbitrary chosen presentations of isomorphic fields. It does not say that the outer universe cannot distinguish the inner one once extra structure is named: the rest of this part shows that birthday does.

Proposition 14.3 (Cross-model coherence; source 09). *Let κ be an uncountable cardinal of both models. There are actual inclusions*

$$\mathcal{B}_\kappa^M \subseteq \mathcal{B}_\kappa^N, \quad H_\kappa^M \subseteq H_\kappa^N,$$

compatible with the interpretations and their definable isomorphisms. Consequently

$$\mathcal{B}_\kappa^M \prec \mathcal{B}_\kappa^N \iff H_\kappa^M \prec H_\kappa^N.$$

Proof. A sign sequence of M remains the same sign sequence in N ; order and birthday are absolute. A small arithmetic certificate existing in M remains valid in N , so the field operations agree. An old set of hereditary size below κ comes with an old bijection witnessing that bound; since κ is still a cardinal, it belongs to H_κ^N .

For code validity, one must not claim that the two models have the same subsets of the coding ordinal. Instead, a relation well founded in M has an ordinal rank function in M , which remains decreasing in N . Its well-foundedness therefore persists despite the extra subsets in N . Root generation says that every node is connected to the root by a finite downward path, as established in [lemma 5.1](#); these paths are absolute. Extensionality is bounded and absolute. Failure of any of the code conditions in M also persists, using its old witness. Thus validity on old codes is unchanged.

The old Mostowski collapse still satisfies its bounded defining equations in N and hence is the same collapse. Equality and membership of decoded old sets are unchanged. Witnesses to their equality and membership can be constructed in M from those collapses, so their required graph codes already exist there. The reverse sign-graph interpretation likewise sends old objects to their actual images. Applying each interpretation to an elementary inclusion now proves the equivalence. $\square$

Unlike two cutoffs in a single universe, $\mathcal{B}_\kappa^M$ and $\mathcal{B}_\kappa^N$ may have different birthday fibers even at the same small ordinal. Their ordered field reducts nevertheless always include elementarily ([proposition 14.2](#)).

15 The preservation criterion and the first new birthday

Theorem 15.1 (Exact preservation criterion; sources 02, 06, 07, 09). *For every common ordinal γ ,*

$$(\mathbf{No}_{<\gamma})^M = (\mathbf{No}_{<\gamma})^N \iff \forall \alpha < \gamma \ \mathcal{P}(\alpha)^M = \mathcal{P}(\alpha)^N. \quad (15.1)$$

If $\gamma = \lambda$ is an epsilon number, these conditions are equivalent to the literal equality $\mathcal{B}_\lambda^M = \mathcal{B}_\lambda^N$ of the canonically presented structures, including all operations and the birthday function. If $\gamma = \kappa$ is an uncountable cardinal in both models, they are also equivalent to $H_\kappa^M = H_\kappa^N$.

Proof. Taking plus positions is an absolute bijection between sign sequences of length α and subsets of α ; this gives (15.1). Once the carriers agree, [proposition 14.2](#) gives equality of all operations; conversely, equal structures have equal carriers.

For H_κ : a subset of $\alpha < \kappa$ has hereditary size below κ in either model, so $H_\kappa^M = H_\kappa^N$ implies that no bounded subset is new. Conversely assume no new bounded subsets. Old sets of hereditary size below κ stay so in N ([proposition 14.3](#)). For $x \in H_\kappa^N$ take in N a valid code (δ, r, t) with $\delta < \kappa$. Its relation word has length $\Lambda(\delta) < \kappa$, so it is old, as are δ and t . The coded relation is well founded in N , hence in M (an old nonempty subset without a minimal element would also be one in N); extensionality is absolute; the collapse computed in M satisfies its recursion in N , so it is the collapse there. Hence $x \in M$, with hereditary size below κ witnessed in M . $\square$

Source 07 proves (15.1) for every common ordinal, source 02 the epsilon-number clause, source 06 the cardinal clause with H_κ , and source 09 the structure equality inside [theorem 16.4](#). The right

side of (15.1) must *not* be replaced by “no new short sequences of ordinals”: a short sequence with unbounded range can be new while every bounded subset below a large cutoff stays old, which is exactly what the Prikry example of section 17 exploits. Nor is “adds no real” identified with “adds no surreal” (example 15.7).

Corollary 15.2 (Countable cutoffs detect exactly new reals; source 02). *If λ is a countable epsilon number, then $\mathcal{B}_\lambda^M = \mathcal{B}_\lambda^N \iff \mathcal{P}(\omega)^M = \mathcal{P}(\omega)^N$.*

Proof. For infinite $\alpha < \lambda$ a bijection between ω and α lies in M and remains one in N , so a new subset of α gives a new subset of ω ; finite ordinals have no new subsets; and $\omega < \lambda$. $\square$

Lemma 15.3 (New sets imply a new ordinal subset; sources 05, 07). *If $M \subsetneq N$, then N has a subset of some ordinal that is not in M .*

Proof. Suppose not, and let $a \in N$. In N enumerate $\text{TC}(\{a\})$ by an ordinal and code the pulled-back membership relation by a subset of an ordinal through an old injective pairing (for example the natural-arithmetic rectangle map). By hypothesis the code, hence the relation, is in M . Well-founded in N , it is well founded in M ; its decoration computed in M is the decoration in N by uniqueness of recursion, and its root value is a . So $a \in M$, and $M = N$. $\square$

Theorem 15.4 (First new birthday; sources 02, 05, 07). *Let $M \subsetneq N$ and*

$$\beta = \beta(M, N) = \min\{\alpha : \mathcal{P}(\alpha)^N \neq \mathcal{P}(\alpha)^M\}.$$

Then β exists and:

- (i) *every surreal of N of birthday below β belongs to M ;*
- (ii) *N has a surreal of birthday exactly β not in M , all of whose proper prefixes are in M ; thus β is the least birthday of an element of $\mathbf{No}^N \setminus \mathbf{No}^M$;*
- (iii) *β is an infinite cardinal of M .*

Proof. β exists by lemma 15.3. A sign sequence of birthday α is determined by α and its plus set, which is old when $\alpha < \beta$; this proves (i), and a new subset of β gives the new sign of (ii), whose proper prefixes are old by (i). No finite ordinal has a new subset. If β were not an M -cardinal, M would have a bijection $f : \nu \rightarrow \beta$ with $\nu < \beta$; for a new $A \subseteq \beta$ the pullback $f^{-1}[A]$ would be old by minimality, and then $A = f[f^{-1}[A]]$ would be old. $\square$

Remark 15.5 (Source 02’s version). Source 02 proved the theorem for a set-forcing extension $N = M[G] \neq M$, obtaining existence of β from the generic filter: a ground-model well-order of the forcing set turns G into a new subset of an ordinal. Sources 05 and 07 remove the forcing hypothesis by lemma 15.3. No claim that all cardinals are preserved is needed. The theorem does not assert that β is regular, and it does not identify the signs of birthday at most ω with real numbers.

Example 15.6 (Adding a real). If $N = M[g]$ adds a Cohen subset g of ω , no finite subset is new and g is, so $\beta = \omega$. The sign with plus set g is a new surreal of birthday ω , and each of its finite prefixes is old. Such an extension changes $\mathbf{No}_{<\kappa}$ for every common uncountable cardinal κ .

Example 15.7 (No new reals, but new surreals; sources 05, 07). Force over M with countable partial functions from ω_1^M to 2, ordered by extension. The union of a descending ω -chain is a condition, so the forcing is countably closed; the successive-decision argument shows that it adds no new countable sequence of ordinals, hence no subset of any old countable ordinal, and preserves ω_1 . The generic union differs from every old subset of ω_1^M by a density argument [13]. Hence $\beta = \omega_1^M$:

all countable-birthday surreals are unchanged, there are new signs of birthday ω_1^M , and the full birthday inclusion still fails elementarity ([theorem 16.5](#)). Agreement on the reals, or even on every countable sign sequence, does not exhaust what the birthday expansion remembers.

Remark 15.8 (The sibling report). The sibling report `foundations-and-computation/surreal-fields-across-universes` works in the same setting of transitive $M \subseteq N$ with the same ordinals and uses the same letter β for the first new birthday; it writes δ for the first new ordinal-sequence length, which is not named here. It studies the external saturation of the *pure* ordered field $\mathbf{No}^M$ inside N , which is elementary in $\mathbf{No}^N$ by [proposition 14.2](#); the present report studies the birthday expansion, whose inclusion is non-elementary as soon as $M \neq N$ ([theorem 16.5](#)). The first-new-birthday theorem and the absoluteness of old arithmetic are shared ground; the countably closed and Prikry examples appear in both.

16 Detecting new subsets

16.1 The old-power-set detector at a cutoff

Theorem 16.1 (Uniform detector for an added subset; source 09). *Let κ be an uncountable cardinal of both models. Suppose $\alpha < \kappa$, N has a subset of α not in M , and*

$$\nu = (2^{|\alpha|})^M < \kappa.$$

Then $\mathcal{B}_\kappa^M \not\prec \mathcal{B}_\kappa^N$. One fixed interpreted formula witnesses the failure using parameters coding α and $A = \mathcal{P}^M(\alpha)$.

Proof. The cardinal bound gives $A \in H_\kappa^M$. In the membership language consider

$$\exists y [\forall v \in y (v \in \alpha) \wedge y \notin A]. \quad (16.1)$$

This is false in H_κ^M , because all its subsets of α lie in A . It is true in H_κ^N : any newly added subset of α has hereditary size below the still-cardinal κ and is not an element of the old set A . Translate (16.1) through $\mathcal{I}$ and choose old codes for the two parameters. Cross-model coherence gives the claimed failure in the birthday structures. $\square$

The membership-language formula (16.1) is Σ_1 : its matrix has only bounded quantifiers. Its fully expanded birthday-field translation need *not* be existential or Σ_1 in that different language, because code validity includes the bit-quantified well-foundedness test. No quantifier-complexity claim is made across that translation.

Proposition 16.2 (Sharpness of the detector's parameter cost; source 09). *The old parameter $\mathcal{P}^M(\alpha)$ belongs to H_κ^M exactly when $(2^{|\alpha|})^M < \kappa$. In the rooted coding scheme, every code for that parameter has node-domain cardinality $|\text{TC}(\{\mathcal{P}^M(\alpha)\})|^M$, which equals the power-set cardinal for infinite α . In that infinite case, a code of birthday cardinality no larger than that cardinal can be chosen.*

Proof. The first statement is [lemma 10.6](#) inside M . The exact node-domain cardinality is (5.7). For infinite α , the transitive closure consists of the power set, its subsets, and the ordinal data beneath α , giving cardinality $2^{|\alpha|}$. Choosing its initial ordinal as the node domain gives relation birthday $\Lambda(\nu)$, of cardinality ν by [lemma 4.3](#). $\square$

This is not a theorem that elementarity must hold when the bound fails. Another formula can detect a change at a smaller cutoff. It says exactly when this particular, universally applicable old-power-set witness is available.

Example 16.3 (A conditional continuum-sized illustration). Suppose M satisfies CH, N preserves ω_2^M , and N adds a real. Then the old set of reals has hereditary size ω_1^M , so the detector works at the cutoff ω_2^M . That old set is not an element of $H_{\omega_1}^M$, so the same parameter cannot be used at the cutoff ω_1^M . No claim of elementarity at the smaller cutoff follows.

16.2 Rigidity under extensions at a strong limit

Theorem 16.4 (Strong-limit extension rigidity; source 09). *Let κ be an uncountable cardinal of both models, strong limit in M . Then*

$$\begin{aligned} \mathcal{B}_\kappa^M \prec \mathcal{B}_\kappa^N &\iff \mathcal{B}_\kappa^M = \mathcal{B}_\kappa^N \\ &\iff \mathcal{P}^M(\alpha) = \mathcal{P}^N(\alpha) \text{ for every } \alpha < \kappa. \end{aligned}$$

Proof. If some subset of an $\alpha < \kappa$ is new, the strong-limit property makes its old power-set parameter small enough, and [theorem 16.1](#) excludes elementarity. Conversely, if all these power sets agree, the structures are equal by [theorem 15.1](#), hence the inclusion is elementary. Equality of the scalar domains forces equality of the corresponding plus-sets, proving the remaining direction. $\square$

This proves [theorem 1.4](#). It applies, for example, to any forcing extension satisfying the stated cardinal-preservation hypotheses, but uses no particular forcing construction. It is a theorem about *standard cutoff structures at a fixed common cardinal*, not a prohibition on abstract elementary extensions supplied by model theory. Source 02 explicitly declined to claim that a proper enlargement at a fixed cutoff is non-elementary merely because it is proper; the theorem above and [theorem 16.1](#) give sufficient conditions, and nothing more is claimed. Their ordered field reducts still include elementarily even when the birthday expansion fails the detector; a set-theoretic change can be invisible to ordered-field formulas yet visible to one birthday-field formula with old parameters.

16.3 The full class, and an explicit row-search formula

For the full classes write $\mathcal{B}^M = (\mathbf{No}^M; 0, 1, +, -, \cdot, <, b)$ and likewise $\mathcal{B}^N$; by [proposition 14.2](#) the old signs form a substructure.

Theorem 16.5 (An outer universe is detectable; sources 05, 07). $\mathcal{B}^M \prec \mathcal{B}^N \iff M = N$. *When $M \neq N$ the failure is witnessed with old parameters: by the $\mathcal{I}$ -translation of (16.1) with codes for α and $\mathcal{P}(\alpha)^M$, for any α with a new subset (source 07), and by the formula New of [theorem 16.6](#) (source 05).*

Proof. If $M \neq N$, [lemma 15.3](#) gives an ordinal α with a new subset. Old valid codes remain valid in N and decode to the same sets (as in [proposition 14.3](#), without a cardinal bound), and (16.1) is false in M and true in N with those parameters. $\square$

Source 05's detector reads the birthday language directly. Fix a nonzero ordinal θ and in M an enumeration $(A_j)_{j < \nu}$, $\nu > 0$, of $\mathcal{P}(\theta)^M$ (a surjection suffices), and code it by one old sign m of birthday $\Lambda(\theta + \nu + 1)$ whose bit at $p_{\theta+\nu+1}(j, \xi)$ is plus exactly when $\xi \in A_j$. Let

$$\text{New}(m, \theta, \nu) : \exists x \left[b(x) = \theta \wedge \forall j < \nu \exists \xi < \theta \neg (\text{Bit}(x, \xi) \leftrightarrow \text{Bit}(m, p_{\theta+\nu+1}(j, \xi))) \right],$$

together with the ordinal and length conditions on θ, ν, m ; it asks for a row of length θ different from every old row.

Theorem 16.6 (An explicit new-subset detector; source 05). *With such parameters, $\mathcal{B}^M \models \neg \text{New}(m, \theta, \nu)$, and $\mathcal{B}^N \models \text{New}(m, \theta, \nu)$ if and only if $\mathcal{P}(\theta)^N \neq \mathcal{P}(\theta)^M$.*

Proof. Every sign of birthday θ in M has an old plus set, which is some row A_j . The old sign m has the same bits in N , so it still has exactly the old rows. A new subset of θ gives a witness in N ; conversely a witness has a plus set differing from every old subset, hence new. $\square$

Source 05 writes the matrix with its rectangle map and length $\theta(\nu + 1)$; the formula is the same up to that pairing. The enumeration is a set in M and the matrix one sign of set length; no class of ground-model signs is named. The formula is not claimed to be existential after the abbreviations are expanded, since the prefix and bit predicates have quantifiers of their own. The detector branches off after the pairing lemma and does not need the arithmetic certificates.

Remark 16.7 (The bounded version). At a cutoff the matrix must fit. Source 05 states: if a common ordinal κ is an uncountable regular cardinal in both models, the old matrix has $\theta, \nu < \kappa$ and length below κ , and N has a new subset of θ , then $\mathcal{B}_\kappa^M \not\equiv \mathcal{B}_\kappa^N$, witnessed by New . Its proof uses regularity only through closure of natural arithmetic below κ , which holds at every uncountable cardinal (lemma 2.2); so the conclusion holds whenever κ is an uncountable cardinal of both models and $(2^{|\theta|})^M < \kappa$, the hypothesis of theorem 16.1. Without a bound on the old enumeration the corollary cannot be inferred from $\theta < \kappa$ alone, because the set of old subsets of a small ordinal may have cardinality at least κ .

Remark 16.8 (The conclusions use parameters). These theorems say that the *canonical inclusion* is not elementary. They do not by themselves show that the two structures have different parameter-free complete theories, nor do they exclude unrelated embeddings or isomorphisms. If the extension changes a parameter-free set-theoretic sentence, its translation does distinguish the theories (sources 05, 07).

17 Prikry forcing: an unchanged field with a singularized cutoff

17.1 The imported forcing theorem

We use the classical ordinary Prikry forcing associated with a normal measure U on a measurable cardinal κ . Conditions consist of a finite increasing stem $s \subset \kappa$ and a measure-one set $A \in U$ above that stem; an extension may lengthen the stem using points of A and shrink A , a direct extension only shrinks A . Its standard properties are: the generic stem union is cofinal of order type ω in κ ; the forcing adds no bounded subsets of κ ; and it preserves cardinals. These are imported from the exposition of Poveda–Rinot–Sinapova, especially Lemma 2.10 and Section 3.1 [11], and are not reproved in full. For orientation, source 02 sketches the bounded-subset consequence of the Prikry property: decide a name for a subset of $\alpha < \kappa$ by successive direct extensions, using that intersections of fewer than κ measure-one sets have measure one; bounded subset preservation keeps the cardinals below κ and κ itself, and the κ^+ -chain condition the cardinals above.

Theorem 17.1 (Forcing invisibility, conditional; sources 02, 06). *Suppose a transitive model M of ZFC has a measurable cardinal κ , and let $N = M[G]$ be an ordinary Prikry extension at κ . Then*

$$\mathcal{B}_\kappa^M = \mathcal{B}_\kappa^N, \quad H_\kappa^M = H_\kappa^N = V_\kappa^M,$$

with the same carrier, functions and relations, although

$$M \models \text{cf}(\kappa) = \kappa, \quad N \models \text{cf}(\kappa) = \omega.$$

The common membership structure satisfies ZFC in both universes.

Proof. In M the measurable κ is strongly inaccessible, hence an epsilon number by lemma 2.2; it remains a cardinal, hence an epsilon number, in N , although singular there. Prikry forcing adds no subset of any $\alpha < \kappa$, so theorem 15.1 gives both equalities of structures. Since κ is inaccessible in M , $H_\kappa^M = V_\kappa^M$ (sets in V_κ have small transitive closures by strong limit and regularity, and a well-founded relation on fewer than κ nodes has rank below κ), and $V_\kappa^M \models \text{ZFC}$. The generic sequence witnesses $\text{cf}^N(\kappa) = \omega$. Truth of each fixed first-order formula in one unchanged set structure is absolute between transitive models, by induction on formulas. $\square$

Corollary 17.2 (No uniform regularity detector; sources 02, 06). *Relative to the existence of such a pair of models, no first-order sentence in the language of $\mathcal{B}_\kappa$ holds, in every universe and at every uncountable cardinal cutoff, exactly when the external cutoff is regular, or exactly when it is strongly inaccessible. More strongly, no invariant determined by the isomorphism type of the enriched structure alone determines the external cofinality of its cutoff in a universe-independent way.*

Proof. Such a sentence or invariant would take different values on the two identical structures of theorem 17.1. $\square$

This is a relative-consistency separation using a measurable cardinal: no measurable cardinal is constructed or proved to exist, and ZFC is not claimed to prove the existence of such a pair of universes. The statement concerns fixed first-order sentences; unrestricted second-order satisfaction is *not* asserted to be unchanged, since the family of external subsets of the carrier changes. The equality established in N is with the *old* V_κ^M ; it is not claimed that $H_\kappa^N = V_\kappa^N$, and the rank hierarchy of the extension can contain new objects. The corollary also shows why diagnosing singularity by failure of some unspecified Replacement instance is unreliable: the small universe keeps every axiom of ZFC while the ambient universe acquires an unbounded sequence invisible to it.

Proposition 17.3 (Where the missing sequence lives; sources 02, 06). *In the situation of theorem 17.1, the first new birthday is exactly $\beta(M, N) = \kappa$. The Prikry sequence $h : \omega \rightarrow \kappa$, and its range, have hereditary size κ , so they lie in $H_{\kappa^+}^N \setminus H_\kappa^N$; any faithful pointed code for them needs at least κ nodes. For every epsilon number λ with $\kappa < \lambda < \kappa^+$, of which there are some, the carrier $\mathbf{No}_{<\lambda}$ changes and its interpreted universe H_{κ^+} contains the new sequence.*

Proof. No subset below κ is new, while the range of h is a new subset of κ (already by regularity of κ in M); theorem 15.4 gives $\beta = \kappa$, and every $\lambda > \kappa$ contains that sign layer. The values of h are cofinal ordinals below κ , so the transitive closure of h contains all of κ and has size κ . For $\kappa < \lambda < \kappa^+$ the cardinal ceiling is κ^+ in both models. Iterating $\alpha \mapsto \omega^\alpha$ ω times from above κ gives a fixed point of cardinality κ , an epsilon number in the interval. $\square$

The proposition distinguishes the *appearance of new elements* from a *change of the complete first-order theory*; it proves the former and does not assert the latter for every enlarged cutoff. By contrast, an extension adding a real changes $\mathbf{No}_{<\kappa}$ at every common uncountable cardinal (example 15.6); the preservation criterion distinguishes the two kinds of extension without referring

to the names of their forcing notions. The surcomplex enrichment is likewise unchanged in the Prikry situation ([theorem 19.8](#)).

Part IV

The full class and the surcomplex numbers

18 The full class: the announced theorem and Kunen rigidity

18.1 The announced global bi-interpretation

Theorem 18.1 (Chen–Hamkins–Yang, announced [[6](#), [7](#), [8](#)]). *In ZFC there are parameter-free first-order interpretations $\mathcal{I}$ and $\mathcal{J}$ such that, formula by formula, $\mathcal{I}(\mathcal{B})$ is definably identified with $(V, \in)$, $\mathcal{J}(V, \in)$ with $\mathcal{B}$, and both composites have definable isomorphisms to the original structures.*

Status and credit. This is the global theorem announced by Chen, Hamkins and Yang, stated there for the surreal field with birthday order; by [lemma 2.1](#) that language and the birthday-function language are definitionally equivalent. Their slides describe the coding of sets by pointed well-founded extensional relations and then by sign sequences, and mark the work as still being checked; the collection’s foundations report records the announcement in `found:sub:announcement`. The written argument below is source 07’s; source 05 gives the same statement with its bisimulation codes, and internally in every model of ZFC ([remark 7.4](#)). It is printed as *a proof of the announced theorem*, not as a new result, and it is not an audit of an unpublished manuscript of Chen–Hamkins–Yang or of their proposed axiomatization.

Proof (source 07). Use the codes of [section 5](#) without a cutoff. Every set has a valid code: choose an injection of $\text{TC}(\{a\})$ into an ordinal (one application of Choice) and code the transported membership relation. The correctness of $\sim$ and $\in_{\mathcal{I}}$ ([lemma 5.2](#)) and of the collapse is proved for set-sized graphs and applies to each code. The reverse interpretation takes sign names $\langle \alpha, A \rangle$ with $A \subseteq \alpha$ and defines the arithmetic by the certificates of [section 3](#), which now exist without a size restriction. The two composites are the collapse relation and the sign-name relation $x \mapsto \langle b(x), A_x \rangle$ of [sections 6](#) and [7](#); both are single formulas, and their correctness is proved one formula at a time, with no truth predicate for V and no class of chosen representatives. $\square$

The local refinements proved in [Part I](#) were stated for V by the announcement; the cutoff theorems, the noncardinal pointed version, and the local certificates at singular cutoffs are what sources 02, 05, 06, 07 and 09 propose to add, with the priority caveats of [section 20](#). Source 09 already noted that its refinements may overlap unpublished details of the Chen–Hamkins–Yang project.

18.2 Transfer of Kunen’s inconsistency

An automorphism and a non-surjective elementary self-embedding must not be conflated. For automorphisms of the full class the argument of [lemma 19.3](#) applies (the ordinals form a definable well-ordered class). It fails for a non-surjective elementary self-embedding j , whose restriction to the ordinals need not be an automorphism, and which satisfies $b(j(x)) = j(b(x))$ rather than the

externally birthday-fixed equation $b(j(x)) = b(x)$. Literature on birthday-fixed surreal morphisms, such as the structures of Rangel–Mariano [16], must be compared with this distinction in mind (source 07).

Lemma 18.2 (Transferring an elementary embedding; source 07). *Let $j : \mathcal{B} \rightarrow \mathcal{B}$ be a class elementary embedding. For each set a choose any code C with $x_C = a$ and put $\hat{j}(a) = x_{j(C)}$, with j applied coordinatewise. This is a well-defined class elementary embedding $(V, \in) \rightarrow (V, \in)$, and $\hat{j}$ is the identity only if j is.*

Proof. Elementarity preserves validity and $\sim$, so the value does not depend on the code, and $\hat{j}$ is a class defined from j without choosing representatives. For each set-theoretic formula and codes for its parameters, theorem 18.1 and elementarity of j give $\varphi(\vec{a}) \iff \varphi(\hat{j}(\vec{a}))$. The definable sign-name composite is natural under elementary maps; if $\hat{j}$ is the identity, it identifies $j(x)$ and x with the same interpreted sign name, so $j(x) = x$. $\square$

The class-theoretic hypothesis. The classical Kunen inconsistency rules out a nonidentity elementary embedding $V \rightarrow V$ under Choice, in the form that permits the embedding as a class parameter in Separation and Replacement (Hamkins–Kirmayer–Perlmutter [12]). We work, as source 07 does, in GBC for convenience; global choice is stronger than needed. The class $\hat{j}$ is definable from j and inherits availability as a class parameter. Elementarity may be read formula by formula; for a single class formulation one may use Δ_0 -elementarity with membership-cofinality, which holds because $\hat{j}(\alpha) \geq \alpha$ and $\hat{j}(V_\alpha) = V_{\hat{j}(\alpha)}$.

Theorem 18.3 (No nontrivial class elementary self-embedding; source 07). *In GBC, or under the class-parameter Separation and Replacement hypotheses just described, every class elementary self-embedding of $\mathcal{B}$ is the identity.*

Proof. A nonidentity j would give a nonidentity elementary $\hat{j} : V \rightarrow V$ by lemma 18.2, contradicting the cited Kunen theorem. $\square$

The set-theoretic input is the classical Kunen inconsistency; this is not a new proof of it. The theorem concerns the full birthday-expanded class. It does not assert that the pure surreal ordered field has no nontrivial embeddings or automorphisms; it does not classify the elementary self-embeddings of the bounded $\mathcal{B}_\kappa$ (which theorem 9.10 reduces to those of H_κ); and it does not prohibit external self-maps of a countable nonstandard model’s internal surreal structure, which need not be classes of that model with the required Separation and Replacement properties.

19 Surcomplex numbers and the orientation obstruction

19.1 The enriched surcomplex structures

For an epsilon number λ , $\mathbf{No}_{<\lambda}[i]$ is algebraically closed because $\mathbf{No}_{<\lambda}$ is real closed. Let $c(x + iy) = x - iy$ be conjugation and

$$\text{cb}(x + iy) = \max(b(x), b(y)),$$

the *coordinate birthday*, whose value is an embedded real ordinal. Put

$$\mathcal{C}_\lambda = (\mathbf{No}_{<\lambda}[i]; 0, 1, +, -, \cdot, c, \text{cb}), \quad \mathcal{C}_\lambda^i = (\mathcal{C}_\lambda, i),$$

and $\mathcal{C}$, $\mathcal{C}^i$ for the full classes on $\mathbf{No}[i]$. The coordinate birthday is a stated convention, not the birthday of a combinatorial-game representative of a complex object, and not asserted to be a pre-existing universal definition of a “surcomplex birthday”. Conjugation, cb and i are explicit extra structure; none is asserted to be definable in the pure algebraically closed field, whose inclusions are elementary in the pure language [14].

Lemma 19.1 (Independence of orientation; source 06). *cb is unchanged if i is replaced by $-i$, and its restriction to $\mathbf{No}_{<\lambda}$ is b .*

Proof. The replacement changes coordinates (x, y) to $(x, -y)$, and negation reverses every sign without changing length, so $\text{b}(-y) = \text{b}(y)$; also $\text{b}(0) = 0$. $\square$

Proposition 19.2 (Recovery of the real birthday structure; all five sources). *Both $\mathcal{C}_\lambda$ and $\mathcal{C}_\lambda^i$ interpret $\mathcal{B}_\lambda$ without parameters on the fixed field of c , with order*

$$x < y \iff \exists t [c(t) = t \wedge t \neq 0 \wedge t^2 = y - x]$$

for real x, y . The structure $\mathcal{C}_\lambda^i$ is uniformly bi-interpretable with $\mathcal{B}_\lambda$, hence with $(H_\lambda, \in)$ at cardinal cutoffs and with $(H_{\kappa(\lambda)}, \in, \lambda)$ otherwise; the same holds for $\mathcal{C}^i$, $\mathcal{B}$ and $(V, \in)$ formula by formula.

Proof. The fixed field is $\mathbf{No}_{<\lambda}$, real closed, where positivity is being a nonzero square; field operations restrict, and cb restricts to b . Conversely interpret complex numbers as pairs of reals with $(x, y)(u, v) = (xu - yv, xv + yu)$, $c(x, y) = (x, -y)$, $i = (0, 1)$ and $\text{cb}(x, y) = (\max(\text{b } x, \text{b } y), 0)$. With i named, $(x, y) \mapsto x + iy$ is definable with inverse

$$z \mapsto \left(\frac{z + c(z)}{2}, \frac{z - c(z)}{2i} \right),$$

and the real composite is $x \mapsto (x, 0)$. These are the definable isomorphisms of a bi-interpretation; compose with [theorem 6.3](#) and [theorem 18.1](#). $\square$

Thus cardinal rounding, recovery of a noncardinal cutoff, and all the set-theoretic distinctions above transfer to the oriented surcomplex structures, and adjoining i creates no new set-theoretic size requirement. The real-axis recovery is standard quadratic-extension algebra; its role here is to identify which enrichment carries the foundational information. In the Prikry situation of [theorem 17.1](#), $\mathcal{C}_\kappa$ and $\mathcal{C}_\kappa^i$ are also unchanged, because both coordinates and all displayed operations are; so adjoining i and retaining conjugation does not repair the inability to recover the ambient cofinality (sources 02, 06).

19.2 Rigidity and the two-element automorphism group

Lemma 19.3 (Rigidity of a standard birthday cutoff; sources 05, 06, 07, 09). *Every automorphism of $\mathcal{B}_\lambda$ is the identity, and every automorphism of a transitive structure $(H_\kappa, \in)$, or $(H_\kappa, \in, \lambda)$, is the identity.*

Proof. The embedded ordinals are definable and numerically well ordered, so any order automorphism of this segment fixes every ordinal by transfinite induction; the automorphism therefore fixes all birthdays, preserves the definable bit predicate at each fixed coordinate, and fixes every sign sequence. For H_κ , external well-founded induction gives $h(x) = \{h(y) : y \in x\} = x$; naming an ordinal does not change this. $\square$

The first assertion is a case of the collection’s existing simplicity rigidity: the report `surcomplex/surcomplex-field-automorphisms` shows, in its section `saut:sec:exponential` (subsection “Simplicity restores the canonical surreal numbers”), that a bijection preserving numerical order and prefix simplicity is the identity, and prefix is definable from order and birthday (lemma 4.1). It is printed here once, not as a new result. Rigidity is not pointwise definability: a rigid structure can have elements that are not individually definable without parameters.

Theorem 19.4 (The orientation obstruction; sources 05, 06, 09, after Jeřábek). *For every epsilon number λ , $\text{Aut}(\mathcal{C}_\lambda) = \{\text{id}, c\}$. Consequently $\mathcal{C}_\lambda$ is parameter-free mutually interpretable with, but not parameter-free bi-interpretable with, $(H_\lambda, \in)$ at cardinal cutoffs, or $(H_{\kappa(\lambda)}, \in, \lambda)$ otherwise; after naming i the bi-interpretation holds (proposition 19.2).*

Proof. An automorphism preserves the fixed field of c , its definable order and birthday, so its restriction is the identity by lemma 19.3; it sends i to a root of $X^2 + 1$, so it is the identity or conjugation, as in `saut:thm:axis` ($\text{Aut}(F(i)/F) = \{\text{id}, c\}$ for real closed F). Both preserve c and cb (lemma 19.1).

A parameter-free bi-interpretation induces mutually inverse homomorphisms between the automorphism groups: an automorphism preserves the definable interpretation domain and equivalence relation, induces an automorphism of the interpreted structure, and the definable round-trip isomorphism recovers it. The set structure is rigid (lemma 19.3), whereas conjugation is a nonidentity automorphism of $\mathcal{C}_\lambda$. This excludes *every* parameter-free bi-interpretation, not only the coordinate one. Mutual interpretation holds: recover the real birthday structure inside $\mathcal{C}_\lambda$ and then its set universe; in the other direction use the reverse surreal interpretation and the pair field, omitting the constant i . $\square$

The general real/complex obstruction, that real and complex fields with conjugation are mutually interpretable but need a named imaginary unit for a parameter-free bi-interpretation, was explained by Jeřábek in 2021 [9]; only source 06 credited it, and the credit applies to the versions in sources 05 and 09 as well. The group computation is a consequence of classical rigidity and of `saut:thm:axis`, not a new classification of the unrestricted field automorphisms of $\mathbf{No}[i]$ studied in the automorphisms report. The point here is the distinction between mutual interpretability and bi-interpretability, and why an unnamed imaginary unit cannot be treated as harmless notation in the latter claim. The two roots i and $-i$ have the same parameter-free type in $\mathcal{C}_\lambda$; interpreting a rigid universe does not force the interpreting structure to be rigid, and it is the definability of both round trips that transmits rigidity (source 06).

19.3 Two lifts of each real embedding

Theorem 19.5 (Two lifts, and only two; source 06). *Let κ, λ be uncountable cardinals. Every elementary embedding $f : \mathcal{B}_\kappa \rightarrow \mathcal{B}_\lambda$ has exactly two elementary extensions $\mathcal{C}_\kappa \rightarrow \mathcal{C}_\lambda$,*

$$\widehat{f}_\epsilon(x + iy) = f(x) + \epsilon i f(y), \quad \epsilon \in \{1, -1\},$$

relative to reference roots of -1 , and every elementary embedding $\mathcal{C}_\kappa \rightarrow \mathcal{C}_\lambda$ is one of these. After naming the reference imaginary units only the positive lift remains. Hence

$$\text{Emb}_{\text{el}}(\mathcal{C}_\kappa, \mathcal{C}_\lambda) \simeq \text{Emb}_{\text{el}}(H_\kappa, H_\lambda) \times \{1, -1\}, \quad \text{Emb}_{\text{el}}(\mathcal{C}_\kappa^i, \mathcal{C}_\lambda^i) \simeq \text{Emb}_{\text{el}}(H_\kappa, H_\lambda),$$

the signs multiplying under composition, and $\mathcal{C}_\kappa^i \prec \mathcal{C}_\lambda^i \iff H_\kappa \prec H_\lambda$ for $\kappa < \lambda$ (source 05 proved the last equivalence, and the positive-lift correspondence, at regular cardinals).

Proof. The displayed maps are field embeddings commuting with c and preserving cb , because f preserves order and birthdays, and they differ on i . For elementarity, name i and translate each complex formula into $\mathcal{B}_\kappa$ by coordinates, as in [proposition 19.2](#); elementarity of f gives elementarity of $\hat{f}_1$, and composing with c gives the other lift in the language without i . Conversely an elementary embedding of the complex structures restricts to an elementary embedding of the definable real structures and sends i to $\pm i$, and the field laws force the displayed form. Combine with [theorem 9.10](#). $\square$

At $\kappa = \lambda$ with $f = \text{id}$ this gives $\text{Aut}(\mathcal{C}_\kappa) = \{\text{id}, c\}$ again. The statement concerns the specified birthday-and-conjugation enrichment; it is emphatically not a classification of automorphisms of the pure algebraically closed field, which has far more symmetry (see the automorphisms report). It is a specialization combining the local theorem with coordinate complexification, not a new general Galois principle.

19.4 The exact parameter threshold for orientation

Theorem 19.6 (Orientation threshold; source 06). *Let κ be an uncountable cardinal and $A \subseteq \mathbf{No}_{<\kappa}[i]$ any set of parameters. The following are equivalent:*

- (i) *some root of $X^2 + 1$ is definable in $\mathcal{C}_\kappa$ over A ;*
- (ii) *a linear order of the whole carrier is definable in $\mathcal{C}_\kappa$ over A ;*
- (iii) *A contains a nonreal element.*

In particular neither an imaginary unit nor a linear order is definable over any collection of real parameters, and one nonreal parameter suffices for both.

Proof. If all parameters are real, conjugation fixes them and exchanges the two roots, so neither root is definable; and a definable linear order $\triangleleft$ would be preserved by c , while for nonreal z exactly one of $z \triangleleft c(z)$ and $c(z) \triangleleft z$ holds and c reverses it. If $z \in A$ is nonreal, $v = z - c(z)$ is nonzero and purely imaginary, $-v^2$ is real and positive, and

$$j = \frac{z - c(z)}{\sqrt{-(z - c(z))^2}},$$

with the positive square root of the definable real field, is definable from z and satisfies $j^2 = -1$. The definable coordinates $(\frac{w+c(w)}{2}, \frac{w-c(w)}{2j})$, ordered lexicographically, give a definable linear order of the carrier; it is not a field order. $\square$

Each defining formula uses finitely many parameters; allowing arbitrarily many real parameters does not help because conjugation fixes them all. The proof uses only that $\mathbf{No}_{<\kappa}$ is real closed. A nonreal parameter removes the twofold obstruction to the coordinate round trip, but if it is kept as a named constant, its two real coordinates must be kept in the real interpretation and the corresponding sets in the H_κ interpretation. It is *not* inferred that an arbitrary expansion $(\mathcal{C}_\kappa, z)$ is bi-interpretable with the *unexpanded* H_κ ; naming a root of -1 avoids the issue because its coordinates $(0, \pm 1)$ are already definable.

19.5 The full class, under GBC

Theorem 19.7 (Surcomplex elementary self-embeddings; source 07). *In the class setting of [theorem 18.3](#), the class elementary self-embeddings of $\mathcal{C}$ are exactly the identity and conjugation, and those of $\mathcal{C}^i$ only the identity. In particular every such self-embedding is surjective.*

Proof. An elementary self-embedding of $\mathcal{C}$ restricts to one of the definable real structure $\mathcal{B}$, which is the identity by [theorem 18.3](#); it sends i to $\pm i$, and the coordinates are fixed, so it is the identity or conjugation. Both are automorphisms. Naming i excludes conjugation. $\square$

This is the one surcomplex result of this report that goes beyond the automorphism statements of the automorphisms report: it concerns elementary self-embeddings that are not assumed surjective, of the full structure, under GBC. With [lemma 19.3](#) in place of Kunen’s theorem the same argument gives only the two-element *automorphism* group of the bounded structures. Nothing is asserted about unrestricted field embeddings, and the specified conjugation and coordinate birthday are not claimed to be recoverable from the pure surcomplex field.

19.6 Outer models on the surcomplex side

Theorem 19.8 (The surcomplex outer-model contrast; sources 05, 07). *For transitive $M \subseteq N$ with the same ordinals, the inclusion of the full surcomplex fields is elementary in the pure field language, and also after naming i and conjugation. After additionally naming cb , with or without i , it is elementary exactly when $M = N$. If $M \neq N$, the least coordinate birthday of a new surcomplex number is $\beta(M, N)$.*

Proof. The pure-field statement is [proposition 14.2](#). With i and c named, translate every formula into real coordinates; it becomes an ordered-field formula, so the inclusion is elementary. With cb named the real axis with its birthday is definable, so an elementary inclusion would restrict to $\mathcal{B}^M \prec \mathcal{B}^N$, which forces $M = N$ by [theorem 16.5](#). If $cb(x + iy) < \beta$, both coordinates are old; a new real sign of birthday β is a new surcomplex number of coordinate birthday β . $\square$

The result concerns field and coordinate structure, not complex analysis: it involves no contour integral, topology or total surcomplex exponential, and is not a consequence of an analytic continuation or a valuation argument.

Part V

Scope, questions and provenance

20 What the theorems do and do not imply

20.1 Logical strength versus field strength

Already $\mathcal{B}_{\varepsilon_0}$ recovers H_{ω_1} . Within that membership structure, the natural numbers and all subsets of the natural numbers form the usual full two-sorted structure of second-order arithmetic ([proposition 12.1](#)). The birthday-field interpretation thus includes quantification over all actual subsets of ω , not merely over computable subsets, because the numbers coding those subsets are first-order elements. This does not make set theory definable in the pure ordered-field reduct, where birthday is not definable ([proposition 12.5](#)), and it does not give a decision procedure ([corollary 12.2](#)).

20.2 Four statements that must not be conflated

For a cutoff λ the following can have very different answers:

Question	Exact issue in this report
Is $\mathbf{No}_{<\lambda}$ a real closed field?	Classical epsilon-number cutoff theorem.
Which sets can be decoded?	Exactly $H_{\kappa(\lambda)}$, by rooted relation codes.
Does the decoded universe satisfy ZFC?	At cardinal cutoffs, exactly the worldly condition.
Is an initial inclusion elementary?	Both cutoffs must be cardinals, followed by elementarity of their hereditary-size universes.

An algebraic closure theorem alone does not answer any of the last three questions. Conversely, a transitive set model satisfying ZFC need not arise from a regular inaccessible cutoff ([proposition 11.7](#)), and the same enriched field can sit below a regular and a singular cutoff in two universes ([theorem 17.1](#)). Hereditary size, not rank, is what the birthday structure measures: the recovered universe is H_κ , which in general differs from V_κ ; the two agree, for example, at every worldly κ ([lemma 11.1](#)).

20.3 Choice and quotients

Ordinary Choice is used to enumerate an individual transitive closure and, in some proofs, to enumerate a particular fiber or choose a particular bijection. The interpretation does not choose one representative simultaneously for every set. It keeps all valid codes and quotients by a definable equivalence relation. This is why Global Choice is unnecessary outside [section 18.2](#).

The exact cardinal-rounding conclusion is stated in ZFC. Without Choice, not every transitive closure is well-orderable, and a code on an ordinal cannot represent every set; [theorem 13.2](#) identifies the exact range of the full-class coding in ZF. No choice-free version of the cutoff theorems is claimed, and nothing in the ZFC proofs justifies silently deleting Choice.

20.4 Set models, nonstandard models, and class truth

At a fixed cutoff, the structures in the main theorems are actual set structures with ordinary first-order satisfaction; their interpretations are not hypothetical syntactic models satisfying only a fragment of surreal arithmetic. A nonstandard model of ZFC can internally perform the same constructions, but its coded well-foundedness and its internal cardinal ceiling must be read in that model; an externally ill-founded model is not covered by an external Mostowski-collapse claim merely because it internally believes its code is well founded. Arbitrary nonstandard models of a proposed surreal-arithmetic axiom system need a separate analysis.

For the full class $\mathbf{No}$, one applies the translation to a fixed formula and fixed set parameters. Statements about “all formulas” are metatheoretic schemes. We do not form a set of all class functions, a proper-class membership quotient, or a truth predicate for V inside ZFC. The cutoff results do not require a class theory, a Grothendieck universe, or a large-cardinal existence axiom; GBC is used only in [section 18.2](#) and [theorem 19.7](#), a measurable cardinal only in [section 17](#), and an inaccessible only in [proposition 11.7](#).

20.5 Why uncountability and the epsilon hypothesis matter

For the hereditary-size bi-interpretation at cardinal cutoffs, uncountability ensures that countably sized computation certificates and the ordinary infinite set ω remain below the cutoff. At ω the

finite-birthday surreals form the dyadic ring, not a field, and only the forward interpretation onto the hereditarily finite sets is claimed ([proposition 8.1](#)). For noncardinal cutoffs, the epsilon hypothesis supplies closure of the bounded pairing terms and field operations. Arbitrary ordinal sign cutoffs that fail those closure conditions are not classified; the full birthday-field statement would then need a different language or partial operations. The reconstruction also applies to the canonical birthday function only, and uses the presence of every sign of each length below the cutoff: a small subfield with an inherited birthday may omit most subset codes, and no such generalization is asserted (source 05). For outer models all ordinals are assumed common; models of different heights are a different problem.

20.6 Non-claims of the sources

Every limitation, disclaimer and priority caveat stated by a source is kept. Most appear at their point of use above; this subsection lists them by source.

Across the report. The proofs are written mathematical arguments. They have not been independently refereed and have not been formalized in Lean; no Lean source is supplied. The finite programs check finite analogues only. No named longstanding conjecture is resolved. Priority is not certified by any of the five targeted literature and repository searches; “not found” is a bounded search result, not proof of absence.

Source 02. The full-class interpretation is Chen–Hamkins–Yang’s prior announced work; its contributions are the bounded refinement and its consequences, with priority not certified. Interpreting $H_{\kappa(\lambda)}$ is not interpreting the ambient universe and does not assert $H_{\kappa(\lambda)} \models \text{ZFC}$. The translated Power Set sentence is one axiom; it does not characterize Replacement, regularity or full ZFC. Its own elementary-inclusion conditions were necessary only (the converse is now [theorem 9.3](#)). The results concern concrete externally well-founded structures. Sharpness is intrinsic to the interpretation. The collector recovers an ordinal value, not a surreal of birthday λ . Prikry forcing is imported; no measurable cardinal is constructed; the statement is relative consistency and concerns fixed first-order sentences, not second-order satisfaction. The preservation criterion must not be replaced by “no new short sequences”, and a proper enlargement at a fixed cutoff is not claimed non-elementary merely because it is proper. The onset proposition concerns new elements, not a change of complete theory. In the surcomplex part i , c and the coordinate birthday are explicit structure, not definable in the pure algebraically closed field, and no new automorphism classification is claimed. Only one-way interpretation was claimed at ring cutoffs and by source 02 generally. The Rangel–Mariano program is not settled. The finite tests do not prove the transfinite results.

Source 05. Its package was called candidate-original: not independently refereed or formally verified; no named problem settled; the audit targeted; priority unverified. It omitted the Chen–Hamkins–Yang credit, which is restored here. Classical inputs (sign arithmetic, cardinal closure, well-founded recursion, quantifier elimination, and the simplicity rigidity of the automorphisms report) are credited. Regularity was its uniform hypothesis and it inferred no singular theorem; $H_\kappa \neq V_\kappa$ in general and $\mathcal{P}(\alpha)$ need not be a set in H_κ . The coordinate birthday is a convention, not a game birthday. No sharp product-birthday bound is used. The outer-model theorem needs transitive same-ordinal models; nonstandard models are handled internally. Its non-elementarity is witnessed with parameters, not by distinct parameter-free theories. The first new birthday is

not asserted regular, and signs of birthday at most ω are not identified with reals. The bounded detector needs an enumeration bound. No small-subfield generalization and no different ordinal heights are covered. The embedding correspondence is a reduction, not an existence theorem, and claims no large-cardinal consistency result; exponential automorphisms are untouched, and the collection's exponential faithfulness result is outside its proof dependencies. The detector formula is not claimed to be existential after expansion. Its Python checks are diagnostic only.

Source 06. Chen–Hamkins–Yang is prior work and the central coding idea is theirs; Jeřábek (2021) is prior for the conjugation obstruction; field closure and Prikry are imported. Novelty is provisional; not peer-reviewed or Lean-certified; no named open problem settled. The coordinate birthday is a convention, and the automorphism conclusions do not apply to the pure surcomplex field or classify its automorphisms. The $\sqsupset_\omega$ theorem does not show that every singular κ fails Replacement, and internal ZFC does not imply external inaccessibility. The ZF boundary concerns this construction only, and internal Choice is not the range statement. Prikry gives relative consistency, not a ZFC theorem, and not $H_\kappa^N = V_\kappa^N$. A nonreal parameter does not give bi-interpretability with an unexpanded H_κ . Rigidity is not pointwise definability. Its reduct corollary is not a minimal-language claim. It left open noncardinal cutoffs (answered here for epsilon numbers), ill-founded models of a surreal-arithmetic theory, quantifier elimination for an order-and-simplicity language, and a universal embedding theorem without Global Choice. The finite tests do not verify the theorems.

Source 07. It omitted the Chen–Hamkins–Yang credit and presented the full-class bi-interpretation as a proposed contribution; that is corrected in [theorem 18.1](#), and it had no Jeřábek credit, now added. Its results were proposed contributions without priority certification, refereeing or Lean. It claims no new proof of Kunen's inconsistency, no rigidity of the pure field, no classification of bounded elementary self-embeddings, and no distinct parameter-free theories from non-elementarity; nothing about unrestricted field embeddings. Its plateau theorem is forward only, and its Choice remark says the construction is not a reconstruction of V without Choice. The Kunen transfer needs GBC or class-parameter hypotheses and does not forbid external self-maps of countable nonstandard models. Nonstandard models are handled internally. The reconstruction is a logical representation, not literal membership between chosen representatives; there is no canonical code, no decidability result, and no simple closed sign rules for arithmetic. Rangel–Mariano and Bertram are distinct programs whose categorical obstructions are not refuted. The finite checks are limited.

Source 09. The global result of Chen–Hamkins–Yang, the van den Dries–Ehrlich cutoff theorem with its erratum, and ordinary well-founded coding are prior work; the refinements may overlap unpublished details of the Chen–Hamkins–Yang project; no named conjecture is resolved; no Lean, no peer review. It did not credit Jeřábek; that credit is added. Cardinal rounding is not a rank cutoff. The principles Et and Pack are explicit schemes, not identified with any formulation of SA, and the independence examples are within this family only. The detector bound is sharp only for the old-power-set parameter; no quantifier-complexity claim is made across the translation; failure of the bound does not imply elementarity. Extension rigidity concerns standard cutoff structures and does not forbid abstract elementary extensions. The orientation result is a consequence of classical rigidity, not a new automorphism classification. Choice is used object by object and no ZF version of its theorems was claimed ([theorem 13.2](#) is source 06's). Nonstandard models are internal; no class theory, Grothendieck universe or large cardinal is needed for the cutoff theorems. At ω the cutoff is a ring; non-epsilon cutoffs are not classified. The singular worldly construction is a standard illustration needing an inaccessible.

21 Open questions, re-scoped

Answered inside this report.

- Source 02’s first question, a sufficient criterion for elementary birthday inclusions: answered by source 09 ([theorem 9.3](#), [remark 9.4](#)).
- Source 02’s caution that a two-way theorem must name the cutoff: answered by source 09’s pointed bi-interpretation ([remark 6.5](#)).
- Source 05’s optimal cutoffs at singular cardinals: answered by sources 06 and 07; at noncardinal epsilon numbers: answered by source 09. Source 06’s and source 07’s noncardinal-cutoff questions: answered by source 09 for epsilon numbers.
- Source 09’s fourth target, the choice-free analogue: answered for the full-class graph coding by source 06 ([remark 13.4](#)).

Still open.

Question 21.1 (Cutoffs that are not epsilon numbers). For which ordinals η that are not epsilon numbers, beyond the ring cutoffs of [proposition 8.1](#) (forward only), is there a uniform reconstruction from $\mathbf{No}_{<\eta}$ with its inherited operations, and what exactly is recovered? (Re-scoped from sources 05, 06 and 07.)

Question 21.2 (Minimal signature). How much of $\{0, 1, +, -, \cdot, <, b\}$ can be deleted while retaining a uniform interpretation of hereditary sets? [Corollary 12.6](#) shows that $<$, b and the graph of one ordinal pairing polynomial suffice; whether $<$ and b alone, or addition with birthday, suffice is open, as is whether multiplication is necessary. (Sources 02, 05, 07; partial progress by 06.)

Question 21.3 (Elementary embeddings and explicit criteria). Replace the condition $H_\kappa \prec H_\tau$ in [theorem 9.3](#) by useful explicit hypotheses in interesting families of cardinals (source 09). Classify the elementary self-embeddings of particular bounded $\mathcal{B}_\kappa$; by [theorems 9.10](#) and [19.5](#) this is the corresponding problem for H_κ (source 07).

Question 21.4 (Forcing and the bounded theory). Which changes of the complete bounded theory can occur under forcing that adds new bounded subsets, and how much of the cutoff predicate is first-order recoverable from an appropriately expanded H_κ (source 02)? The detectors of [section 16](#) show failure of elementarity with parameters; they do not decide changes of parameter-free theories.

Question 21.5 (Quantifier complexity). What is the least quantifier complexity of a uniform formula detecting all proper same-ordinal outer-model inclusions in the birthday language (source 05)?

Question 21.6 (Surcomplex reducts). What set-theoretic information is recoverable from $(\mathbf{No}_{<\kappa}[i]; 0, 1, +, -, \cdot, \text{cb})$ when neither conjugation nor i is named (source 05)? [Theorem 19.6](#) concerns the language with conjugation.

Question 21.7 (Axiomatics and formalization). Establish definitional equivalence, or its failure, between Et and Pack and particular published axiom formulations of surreal arithmetic, once finalized (source 09); analyse ill-founded models of such a theory, quantifier elimination for an order-and-simplicity language, and a universal embedding theorem without Global Choice (source 06); and formalize the size-certified interpretation, beginning with the prefix, bit and pairing lemmas, then the certificate verifier and its hereditary-size bound (sources 05, 06, 07, 09).

The collection’s Lean library already contains relevant sign-sequence infrastructure (prefix truncation, the failure of birthday precedence to be prefix simplicity, Hessenberg agreement of ordinal arithmetic), but none of the interpretation theorems of this report is formalized.

22 Provenance and verification

22.1 The sources' repository and literature audits

All five sources inspected the repository read-only at the pinned commit

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and modified nothing. Source 02 inspected the tree, the root and documentation indexes, the catalogue source and the foundations report, and records its scope in `02-birthday-cutoffs-SOURCE_AUDIT.md`, kept byte-identical. Source 05 records its requested line ranges, some truncated, in `data/05-birthdays-recover-sets-repository_audit.json`. Source 06 inspected listings, the first 200 lines of the documentation catalogue, the first 150 lines of the foundations README, and the README and first 240 lines of the automorphisms article. Source 07 inspected the tree, READMEs, catalogue, formalization ledger, the foundations introduction and the automorphisms README. Source 09 inspected the catalogue, the relevant parts of the foundations report and the scope of the automorphisms report. The corrections these audits need are in [section 1.5](#).

The sources' literature checks were: the Chen–Hamkins–Yang announcements and slides (sources 02, 06, 09; source 06 cites slide pages 103, 107 and 122, and source 02 pages 96–122, which are compatible); van den Dries–Ehrlich and its erratum, Bournez–Guilmant, Poveda–Rinot–Sinapova, and Rangel–Mariano (sources 02, 06, 09); Freire–Hamkins, Ehrlich's paper on the simplicity hierarchy [18] (record and abstract only) and Ehrlich–Kaplan [19] (background, not an input) (source 05, whose session could not load the publisher PDF of the erratum); Bertram and Hamkins–Kirmayer–Perlmutter (source 07); Jeřábek (source 06). These external checks were not repeated for this merge.

22.2 Dependency and novelty ledger

Result	Main inputs	Status here
Global birthday-field bi-interpretation (theorem 18.1)	Pointed well-founded relation coding	Prior Chen–Hamkins–Yang announcement; written proof from source 07; not claimed new.
Real-closed epsilon cutoffs	Classical surreal field theory	Prior van den Dries–Ehrlich, with erratum.
Small arithmetic certificates, two routes	Option graphs (09); prefix-pair tables (05, 06, 07)	Explicit local proof ingredients; no priority claim for the general locality principle.
Cardinal rounding and cutoff recovery	Bounded pairing, rooted collapse, collector, sign-graph bridge	Proposed local refinement (09), with cases in 02, 05, 06, 07.
Elementary-inclusion spectrum and embeddings	Collector and coherent bi-interpretations	Proposed classification (09); embeddings (05, 06).
Power Set, eternity, packing, worldly, $\beth_\omega$	Hereditary cardinal bounds	Proposed calibrations (02, 06, 09); the underlying H_κ facts are standard.

Result	Main inputs	Status here
ZF range	Collapse in ZF	Direct refinement of the coding (06); no claim about other interpretations.
Preservation, first new birthday	Sign/subset bijection	Direct consequences (02, 05, 06, 07), tools rather than isolated discoveries.
Detectors, extension rigidity, $M = N$	Old-power-set parameter; row search	Proposed explicit refinements (05, 07, 09); outer-model reasoning is standard.
Prikry invisibility	Imported Prikry forcing	Proposed application under a measurable cardinal (02, 06).
Kunen transfer	Imported Kunen inconsistency (GBC)	Proposed application (07).
Surcomplex orientation, automorphisms	Real-axis recovery, rigidity, Jeřábek	Consequence of classical rigidity and the automorphisms report.
Two lifts, orientation threshold, full-class self-embeddings	Coordinates, rigidity, Kunen	Proved specializations (06, 07); the only surcomplex items new to the collection.

The places where the argument could not be replaced by a general slogan are the arithmetic certificates (saying that the global construction is definable does not prove that the same formula computes arithmetic in an H_κ without Replacement), and the collector (a noncardinal cutoff is not recovered merely by interpreting the unpointed hereditary-size universe).

22.3 Finite checks

Each source supplied a standard-library Python program checking finite analogues of its coding conventions. All five were rerun for this merge with Python 3.14.4 on copies of the files; each passed, and each recorded output was reproduced exactly, apart from line endings. The recorded runs are:

Source	Recorded finite checks
02	65,025 prefix comparisons and 1,538 bit positions through length 7; 173,880 natural-ordinal matrix addresses on 81 Cantor-normal-form samples; 687 relabellings of all 16 sets of V_4 ; 1,024 code pairs for equality and membership; 4 invalid codes rejected.
05	65,025 prefix comparisons through birthday 7, including the counterexample of remark 4.2 ; 64 acyclic graphs on four nodes, 4,096 graph pairs, 65,536 root-pair checks each for bisimulation equality and membership; 64 matrix round trips; 49 rectangle packings; 30 missing-row detector tests.
06	16,129 prefix and 642 bit instances on 127 words of length at most 6; pairing on 10,000 pairs; all 1,570 rooted relations on at most three vertices, 15 valid codes, 225 pairs each for equality and membership.
07	16,129 prefix comparisons, 642 sign checks, 819 radix addresses (3 radices), 42,849 graph pairs each for isomorphism and membership over 207 labeled graphs decoding 12 sets; the missing-range counterexample confirmed.
09	16,129 prefix pairs, 642 bit positions, 11,440 pairing checks; 530 relations and 1,570 rooted relations on at most three nodes, 15 valid codes, 225 pairs each for equality and membership; the counterexample of Appendix B confirmed.

These tests can expose a reversed edge, a missing root or range condition, a bad pairing index, or an incorrect finite sign formula. They do not prove transfinite well-foundedness, cardinal bounds, bi-interpretability, Replacement equivalences, forcing statements, or any large-cardinal assertion; those rest on the written proofs and the explicitly imported classical theorems.

23 Conclusion

Birthday truncation has a precise set-theoretic effect. At an epsilon-number cutoff λ , rooted sign codes recover the hereditary-size universe at the cardinal ceiling of λ ; when λ is not itself a cardinal, its value survives as a definable interpreted set, which explains why noncardinal cutoff inclusions never become elementary even though all their ordered field reducts are elementary subfields. Packing a birthday fiber requires its full power-set cardinal to fit below the cutoff; bounding definable ordinal-indexed birthday growth is Replacement at cardinal cutoffs; together they detect worldly, not merely inaccessible, cardinals. Across universes, the birthday structure below a cardinal is unchanged exactly when no bounded subset is added, while any new set is detected by an explicit formula; yet a measurable cardinal can be singularized without changing the structure at all. The same information is available in surcomplex arithmetic once conjugation, coordinate birthday and an imaginary unit are specified; omitting the unit preserves mutual interpretation but leaves a two-element obstruction to bi-interpretation. The decisive structure is not the field alone: it is the field with a carefully specified memory of how its elements are born.

A Formula expansion and size checks

A.1 What “first-order” means in the coding proof

Every abbreviation expands to a finite formula in $\{0, 1, +, -, \cdot, <, b\}$. Bounded ordinal quantifiers are shorthand for ordinary scalar quantifiers with guards $\text{Ord}(x)$ and $x < \delta$. A set variable in an interpreted membership formula becomes a triple of scalar variables. Equality becomes $\sim$, membership becomes $\in_{\mathcal{I}}$, and quantifiers are relativized to Code:

$$(u = v)^{\mathcal{I}} = (C_u \sim C_v), \quad (u \in v)^{\mathcal{I}} = (C_u \in_{\mathcal{I}} C_v), \quad (\exists u \varphi)^{\mathcal{I}} = \exists \delta_u \exists r_u \exists t_u (\text{Code}(C_u) \wedge \varphi^{\mathcal{I}}),$$

commuting with the connectives. Induction on formulas proves $H_{\kappa(\lambda)} \models \varphi(x_{C_1}, \dots) \iff \mathcal{B}_\lambda \models \varphi^{\mathcal{I}}(C_1, \dots)$ for valid codes, independently of the representatives.

For example, a graph code f on $\delta \times \epsilon$, embedded in the square of width $\theta = \delta + \epsilon + 1$, uses

$$F(i, j) \iff i < \delta \wedge j < \epsilon \wedge \text{Bit}(f, p_\theta(i, j)).$$

It is a total function exactly when

$$\forall i < \delta \exists j < \epsilon (F(i, j) \wedge \forall k < \epsilon (F(i, k) \longrightarrow k = j)).$$

Injectivity is $\forall i, i' < \delta \forall j < \epsilon (F(i, j) \wedge F(i', j) \rightarrow i = i')$, and surjectivity is $\forall j < \epsilon \exists i < \delta F(i, j)$. Occurrences of $f(i)$ in the main text can therefore be eliminated by these unique-value clauses. Root preservation and edge correspondence add only finitely many quantifiers. The graph code’s birthday is $\Lambda(\theta)$; all plus positions outside the relevant rectangle can be required absent, which makes canonical relation coding a definable operation when the intended relation is specified by a formula and its bit string exists.

The dependency order of the abbreviations is

Predicate	Earlier definitions needed	Quantification simulated
$\text{Ord}(a)$	$b(a) = a$	Embedded ordinals
$\text{Pre}(x, y)$	$<, b$	Literal prefix test
$\text{Bit}(s, a)$	$\text{Ord}, \text{Pre}, <, b$	One sign position
E_C	Bit, p_δ	Edges of a code
$\text{Code}(C)$	shape, clean bits, (5.1)–(5.3)	Pointed membership graphs; all subsets of one domain
$\sim, \in_{\mathcal{I}}$	Code, E , coded maps	Interpreted equality and membership
$\text{OC}(a), \text{VisOrd}$	$\text{Ord}, \text{Bit}, p_\theta, \sim$	Literal ordinal as a set code
Coll	Code, $\in_{\mathcal{I}}, \text{VisOrd}$	Collection of literal ordinals

The table follows source 02’s ledger, with this report’s names. A tuple variable is eliminated by replacing it with ordinary variables; “exactly one” has its usual two-quantifier expansion; variables are renamed before substitution to prevent capture. The specification is an ordinary finite formula, albeit a long one.

A.2 Every potentially large construction has a stated bound

Object	Bound used in the proof
Subsets tested in well-foundedness	Individual sign strings of birthday δ ; no collection of that whole fiber is required.
A relation on a code domain δ	Sign birthday $\Lambda(\delta) = \delta^2 + \delta < \lambda$.
An isomorphism or member embedding	Square width $\delta + \epsilon + 1 < \lambda$, with the corresponding Λ bound.
Collapse value of a rooted code	Transitive closure has exactly $ \delta $ members.
An arithmetic computation	At most μ word vertices or table cells, hereditary size μ , where μ bounds the inputs and is infinite.
Rank labels in that computation	Below μ^+ , hence each has cardinality at most μ .
Replacement image after eternity	One bound $\max(\aleph_0, \delta , \varrho) < \kappa$, not an arbitrary singular-cardinal union.
A packed birthday- α fiber	Exists exactly if $2^{ \alpha } < \kappa(\lambda)$.
The old-power-set parameter	Hereditary size $(2^{ \alpha })^M$ for infinite α .
Actual power set of a set of hereditary size μ	At most 2^μ ; this need not be below κ , and no earlier row needs it to be.

B An essential negative regression

In the membership definition, edge preservation alone is insufficient. Take the code for $x = \emptyset$ and the three-node chain coding $y = \{\{\emptyset\}\}$. Map the sole node of the first code to the middle node of the chain, which codes $\{\emptyset\}$. The image root is a child of the target root, and the one-node induced subgraph has no edges, just like the source. But $\emptyset \notin \{\{\emptyset\}\}$. The predecessor-full clause (5.6) rejects this false witness, because the middle node has a predecessor omitted from the image. The finite suites of sources 06, 07 and 09 confirm this counterexample.

C A small concrete membership code

Let $x = \{\emptyset, \{\emptyset\}\}$, the von Neumann ordinal 2 (source 02’s example). Its transitive closure with the root included has the three nodes $0 \leftrightarrow \emptyset$, $1 \leftrightarrow \{\emptyset\}$, $2 \leftrightarrow x$. Take $\delta = 3$, root $t = 2$, and a relation word of length $\Lambda(3) = 12$. The edges are $(0, 1)$, $(0, 2)$ and $(1, 2)$; under $p_3(u, v) = 3u + v$ the plus positions are 1, 2, 5, and all other positions of the length-12 word are minus. Its collapse is the identity on the ordinal 3, with root value 2.

A different labeling can put x at label 0, the empty set at label 2, and $\{\emptyset\}$ at label 1. The relation word then changes, and the root is no longer the largest label. Nevertheless the isomorphism code identifies the two roots, so interpreted equality cannot be literal equality of the triples. The example also shows why label order must not be confused with membership order in the countable-ordinal overflow of [corollary 5.6](#).

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